

LOTTERY DESIGN FOR SCHOOL CHOICE

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ABSTRACT. We use a continuum model of school choice to study student outcomes when priorities are generated by lottery. We consider the case where a single lottery is used to rank all students, and where lotteries are conducted independently at each school. We present three main insights.

First, we compare the number matches in the two systems, and find that the distribution of the length of students' lists plays a key role. If this distribution has an increasing hazard rate, then independent lotteries match more students; if it has a decreasing hazard rate, this reverses. To our knowledge, this is the first work to compare the size of stable matches under different priority rules.

Second, we ask which students prefer a single lottery to independent lotteries. Among students who submit short lists, the rank distribution under a single lottery stochastically dominates the corresponding distribution under independent lotteries. Among students who submit long lists, more match when schools use independent lotteries.

Finally, we study the magnitude of the difference between these procedures. Under independent lotteries, the fraction of assigned students who receive their first choice may be arbitrarily small. We provide a tight lower bound on this quantity when priorities are determined by a single lottery. If schools have a single seat and the most popular schools receive r times the application volume of the least popular ones, this fraction is at least $\sqrt{r}/(1+r)$. Our guarantee increases with the number of seats available at each school.

1. INTRODUCTION

Many cities across the world allow students to choose which public school they will attend. In the United States, 56% of major school districts have a school choice program, up from 29% in 2000 (Whitehurst, 2017). Advocates argue that school choice offers increased equity and efficiency relative to neighborhood assignment: students from neighborhoods with undesirable schools can go elsewhere, and students can select schools with curricula and clubs that match their interests. These advantages are mitigated by capacity constraints, which prevent schools from admitting every student who wishes to attend. As a result, districts must have a policy to determine who gets seats in popular schools.

The Deferred Acceptance (DA) algorithm is an increasingly popular approach, having been adopted in cities across the United States and Europe.¹ This algorithm is inspired by the system for matching doctors to residencies at US hospitals (Roth and Peranson, 1999; Abdulkadiroglu and Sönmez, 2003). One important input is a priority list (ranking of students) for each school, which determines who to reject when too many students express interest.

¹American cities using Deferred Acceptance include New York City (Abdulkadiroglu et al., 2005a), Boston (Abdulkadiroglu et al., 2005b), Chicago (Pathak and Sönmez, 2013), New Orleans (Abdulkadiroglu et al., 2017), Denver, Washington DC, and Newark (Pathak, 2016). Internationally, it is used to in Amsterdam (de Haan et al., 2018), Paris (Fack et al., 2019), and London (Pathak and Sönmez, 2013), as well as for higher-education in many countries (see Table 1 by Fack et al. (2019)).

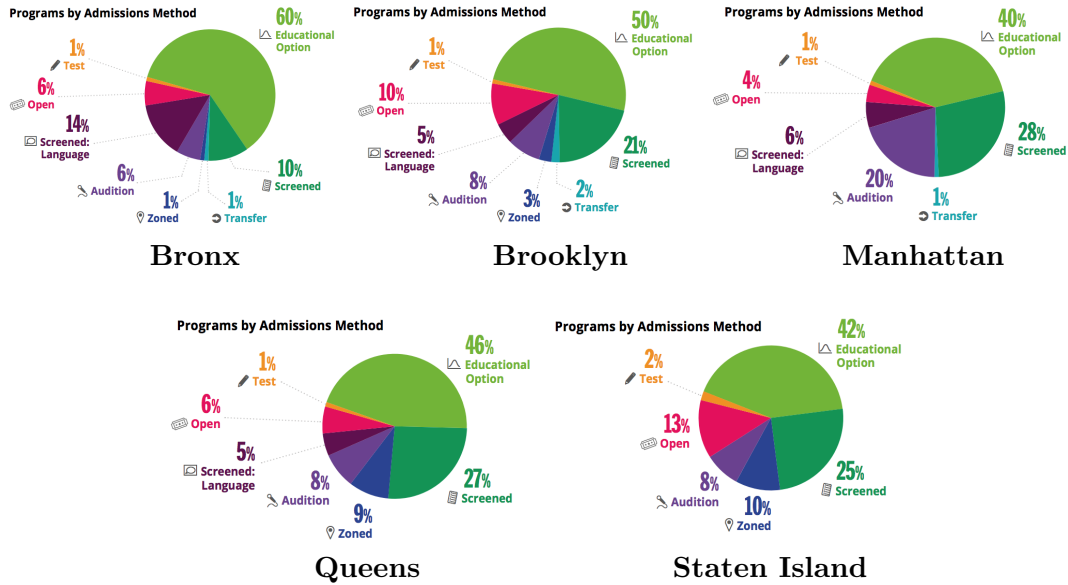


FIGURE 1. Method for determining priorities for programs in each of New York’s five boroughs. “Open” programs use a lottery for all of their seats, while “Educational Option” programs use a lottery for 50% of their seats, and other criteria for the remainder. Screened schools use criteria such as grades, standardized test scores, and attendance. Source: 2019 NYC High School Directory (DOE, 2018).

Methods for determining these priority lists vary widely across cities, schools, and even programs within a school. For Boston elementary schools, the system is as follows. Returning students have the highest priority, followed by students with a sibling already at the school. Next, priority goes to those who live in the region where the school is located (the city is divided into two regions: East Boston and everything else). Because these priorities are coarse and the algorithm needs a complete ordered list, a lottery is used to order students within each priority class (BPS, 2017). For New York City high schools, the system is more complex: each school may consist of several programs, and each program has its own priority list. “Open” programs use a lottery to generate their list. “Zoned” programs guarantee admission to those who live nearby, and arts programs select on the basis of an audition. “Screened” programs rank students on the basis of grades, attendance, and standardized test scores, while “Screened: Language” programs prioritize English language learners. Then there are “Educational Option” schools, which use a lottery to determine priorities for half of their seats while assigning the remainder using a mix of criteria including geography, academic performance, and attendance at the school’s open house (DOE, 2018). Figure 1 shows the frequency of these approaches in each of New York’s five boroughs.

Unsurprisingly, priorities are often contentious. In Boston, a dispute over whether to prioritize students who lived within the “walk zone” for a school resulted in an uneasy compromise in which half of the seats at each school would grant walk zone priority, and half would not (Dur et al., 2018). In New York, specialized high schools have traditionally used a single test score to determine priority. Recent changes save seats for low-income students who score below the admissions

threshold, but are being challenged in court.² In Amsterdam, a lawsuit prompted the school district to change from using separate lotteries at each school to using a single district-wide lottery.³

Despite the importance and contentiousness of school priorities, there are no “best practices” for how they should be generated. In part, this is because the effect of priorities on the final assignment is complex, and poorly understood by academics and practitioners alike. Kojima (2012) demonstrates that increasing the priority for minority students may actually make all minority students worse off! While this result relies on carefully constructed preferences, in practice, giving high priority to certain students may not materially benefit them. For example, Dur et al. (2018) show that the aforementioned Boston walk zone compromise led to outcomes similar to those that would have resulted from ignoring walk zones entirely. When this was discovered, the district reversed its policy and eliminated walk zones, in an effort to be more transparent. This episode highlights the difficulty of trying to achieve policy goals by adjusting priorities whose effects on outcomes are poorly understood. In February 2019, a report from the Brookings Institution (Kasman and Valant, 2019) identified this complexity as a key challenge:

Even if DA algorithms are relatively simple, predicting how student assignment policies will affect enrollment and outcomes is difficult... What happens if the priority applies only to a subset of the seats in a school? What are the consequences of putting it higher or lower in the hierarchy of priorities? These questions are hard to answer without considerable analysis.

One important question frequently facing practitioners is how to break ties. Pathak (2016) remarks that “Questions related to tie-breaking have re-appeared in nearly every city I have interacted with using [Deferred Acceptance].” As mentioned above, ties arise when schools sort students into a small number of groups, and are broken to create a strict ordering by lottery. This can be done with independent lotteries at each school, or with a single lottery used at all schools. Parents often feel intuitively that independent lotteries are “more fair,” but academics have cautioned that they may produce inefficient outcomes ex post (Abdulkadiroglu and Sönmez, 2003; Abdulkadiroglu et al., 2009). Neither argument directly addresses how outcomes differ, or who might benefit from each system. In practice, this question has been tackled by simulation, using data from New York and Boston (Abdulkadiroglu et al., 2009) and Amsterdam (de Haan et al., 2018). Pathak (2016) writes that “as far as I know, there is no stronger ex ante argument for single versus multiple tie-breaking based on the distribution of matchings.”

This paper provides such an argument. We use a continuum model of stable matching to compare outcomes when priorities are common across schools to those when they are generated by independent lotteries. We assume that priorities are fully determined by these lotteries, so the single lottery is equivalent to a random serial dictatorship. Our model allows schools to differ in their capacity and popularity, and students submit lists of varying lengths. We derive expressions for the number of students receiving one of their top- k choices under each procedure, and compare these quantities analytically.

²See a copy of the [court filing](#), as well as [this report](#) by the Washington Post.

³Using independent lotteries produced an inefficient assignment, and [parents sued](#) to be able to exchange school seats. They [lost the case](#), but the district [changed its procedures](#) the following year (all links in Dutch).

We first study the number of unassigned students under each procedure. In practice, many students fail to match to any of the schools on their list: this occurs for 3.8% of students in Boston and 7.1% in New York (Abdulkadiroglu et al., 2009), and 18.7% in New Orleans (EnrollNOLA, 2017). In some cities, these students are administratively assigned to a school with unfilled seats; in others, they are invited to apply to these schools in a second round.⁴ Regardless of which approach is taken, these students are often disappointed with the process, and may leave for a private school, generate negative headlines, or create administrative burdens by challenging the outcome.

We ask which lottery procedure assigns more students to one of the schools on their list. Theorem 1 and Proposition 2 state that the answer depends on the distribution of student list lengths: if it is light-tailed, independent lotteries will assign more students, while the opposite holds if the distribution is heavy-tailed. Interestingly, the comparison does *not* depend on the distribution of popularity and capacity across schools. Applying these results to the list length distributions observed in New York and Amsterdam, the model predicts that independent lotteries will assign more students. This is consistent with the simulation results from Abdulkadiroglu et al. (2009); de Haan et al. (2018). To our knowledge, this is the first paper to analytically compare the number of assigned students under different priority schemes.

Theorem 2 compares the ranks that students assign to their matches. It implies that a single lottery always gives more students their first choice. This prediction also agrees with the aforementioned simulation results. Looking beyond first choices, it states that more students receive one of their top several choices under a single lottery, but that students with the longest lists are always more likely to match under independent lotteries. This adds nuance to the findings on the number of assigned students by explaining *who* benefits from each scheme: a single lottery increases the match rate among students with short lists, at the expense of a lower match rate for students with long lists.

Finally, we study the magnitude of the difference between the two priority schemes, and show that it can be extreme. Theorem 3 provides tight lower bounds on the fraction of assigned students who receive their first choice under a single lottery. This bound depends only on the distribution of popularity and capacity across schools, and not on the number of students, or the length of their lists. Proposition 3 gives a simpler lower bound expressed in terms of the maximum ratio of popularity across schools. For example, if no school is listed more than nine times as often as any other, then at least 30% of students will receive their first choice. If school capacities are large, this bound rises to 50%. Theorem 3 also establishes that no such bound is possible when lotteries are run independently: even if schools are large and equally popular, the number of students receiving their first choice may approach zero as the length of student lists grows.

The remainder of the paper is organized as follows. Section 2 discusses the relationship between this paper and prior work. Section 3 introduces the model. Section 4 presents our main findings, and Section 5 provides intuition for these findings. Section 6 concludes with a discussion of whether students lists react to school priorities, and directions for future work.

⁴New York [is changing](#) from the latter procedure to the former for students entering high school in 2020.

2. RELATED WORK

In their seminal paper, Gale and Shapley (1962) defined the concept of a stable matching, and proved that the Deferred Acceptance algorithm will always find such a matching. Roth (1984) identified this algorithm was being used by the NRMP to match medical residents to hospitals, and Roth and Peranson (1999) describe the redesign of this algorithm. Abdulkadiroglu and Sönmez (2003) proposed using the Deferred Acceptance algorithm to assign seats at public schools, and it has since been adopted in many cities. (Pathak (2016) provides a partial list).

School priorities are often coarse, and group many students into a single tier. This necessitates the use of lotteries to break ties. There are many ways to run these lotteries, and Pathak (2016) writes, “Questions related to tie-breaking have re-appeared in nearly every city I have interacted with.” Erdil and Ergin (2008) show that it is impossible to produce a student-optimal stable matching in a strategy-proof manner, and subsequent papers (and implementations) have focused on the two simple procedures discussed in this paper: running independent lotteries at each school, and running a single lottery to be used across all schools.

The literature on matching is vast: for surveys, see Roth and Sotomayor (1990), Pathak (2011), Abdulkadiroglu and Sonmez (2013) and Kojima (2015). The remainder of this section focuses on work that studies match outcomes when priorities are independent or identical across schools.

2.1. Top Trading Cycles with Lotteries. It is not a priori obvious that the method of breaking ties matters. For example, the Top Trading Cycles algorithm (attributed to David Gale, introduced by Shapley and Scarf (1974), and adapted for school choice by Abdulkadiroglu and Sönmez (2003)) uses the same input as Deferred Acceptance: preferences, priorities, and capacities. It turns out that under TTC, independent lotteries at each school result in an *identical* distribution of outcomes as using a single lottery! This was first observed by Sonmez and Abdulkadiroglu (1998), and has since been extended to more general settings and mechanisms (Sonmez and Unver, 2005; Pathak and Sethuraman, 2011; Lee and Sethuraman, 2011; Carroll, 2014; Ekici, 2015). Most recently, Bade (2019) showed that given any deterministic, strategy-proof, Pareto efficient and non-bossy allocation rule, randomly permuting the roles of agents yields a distribution over allocations that is equivalent to a random serial dictatorship.⁵

2.2. Deferred Acceptance with Lotteries. This equivalence result does not apply to the Deferred Acceptance algorithm, which is bossy and not Pareto efficient. Indeed, under Deferred Acceptance, the lottery method matters. Abdulkadiroglu and Sönmez (2003) observe that using independent lotteries at each school may produce an outcome that is not Pareto efficient. Abdulkadiroglu et al. (2009) show that for any student-optimal stable matching, there exists a single lottery draw that produces it. However, Pathak (2011) notes, “These statements are from an ex post perspective, and there is currently no known stronger ex ante argument for single versus multiple tiebreaking based on the distribution of matchings.”

In place of this argument, Abdulkadiroglu et al. (2009) use simulations based on data from Boston and New York to compare these procedures. They find that a single lottery gives more students

⁵If the district uses TTC with coarse deterministic priorities and lotteries serving only as a tiebreaker, then using a single lottery and using independent lotteries are no longer equivalent.

their first choice (in NYC, the difference is approximately 2,200 students), but leaves more students unassigned. de Haan et al. (2018) observe a similar pattern in Amsterdam. Pathak (2011) writes, “these empirical results raise the need for quantitative results in matching theory that provide guidance on what features of the student preferences and school priorities are responsible for these differences.”

2.3. Quantitative Results in Matching Theory. The first such results came from Pittel (1989). He considered a large one-to-one market with an equal number n of students and schools, complete preferences drawn independently and uniformly at random, and priorities drawn from independent lotteries at each school. He showed that under the student-proposing Deferred Acceptance algorithm, students’ average rank is approximately $\log(n)$. Pittel (1992) shows that under these same assumptions, with high probability every student receives a school ranked no worse than $\log^2(n)$. Ashlagi et al. (2017) relax the assumption that there are equally many students and schools. They show that adding a single student dramatically increases students’ average rank, to $n/\log(n)$.

Furthermore, Ashlagi et al. (2017) study outcomes when using a single lottery, and find that this results in a similar average rank to independent lotteries when there are enough seats for all students, but a much better average rank when there is a shortage of seats. Ashlagi and Nikzad (2019) extend the comparison beyond students’ average rank. They show that when there is a shortage seats, the student rank distribution when using a single lottery “almost stochastically dominates” that which arises from independent lotteries. By contrast, when there are enough seats for all students, the two distributions are incomparable. Ashlagi et al. (2019) study another metric: the fraction of students receiving one of their first k choices. They show that when there is a shortage of seats, this fraction tends to zero as the market grows when using independent lotteries, whereas it remains constant under a single lottery. Collectively, these papers tell a consistent story: when there is a shortage of seats, using a single lottery gives more students one of their top choices, but the comparison is more ambiguous when there are enough seats for all students.

2.4. Our Work in Context. All of the papers mentioned in Section 2.3 assume ex-ante identical schools, and that all students list the same number of schools. This is for tractability, as the stochastic analysis of a finite market quickly becomes complex. To sidestep this complexity, our work uses a formal continuum model. In contrast to the continuum model of Azevedo and Leshno (2016), each of our schools has a finite capacity, and therefore outcomes (i.e. admissions cutoffs) for each school are stochastic. We present a clean characterization of the distribution of student and school outcomes, and the tractability of our model makes it possible to study settings with heterogeneous schools and students. We adopt the preference formation model of Immorlica and Mahdian (2005) and Kojima and Pathak (2009), in which schools are drawn in proportion to their (one dimensional) “popularity.” We place no restrictions on the joint distribution of school popularity and capacity. Furthermore, rather than assuming that all students list the same number of schools, we allow student list length to be drawn from an arbitrary distribution.

We use our model to answer new questions. Abdulkadiroglu et al. (2009) report that 7.1% of high school applicants in New York were not assigned to any school. It is natural to ask how changes in priority might affect this number. To our knowledge, Theorem 1 provides the first

analytical comparison of match size under different priority rules. Most of the aforementioned work assumes complete lists, implying that the number of matches is simply the minimum of the number of students and seats (under any priority rule). Ashlagi et al. (2019) do consider the case where student lists are truncated, and provide upper bounds on the number of unassigned students under independent lotteries. In contrast, our model makes exact predictions for this quantity, and identifies that the list length distribution (assumed to be deterministic in their work) is the key determinant of whether single or independent lotteries assign more students.

Theorem 2 establishes that when students list the same number of schools, the rank distributions under single and independent lotteries are incomparable (by first-order stochastic dominance). This is consistent with the findings of Ashlagi and Nikzad (2019) for the case of uniform random preferences with an excess of seats, but may seem at odds with their finding of almost stochastic dominance when there is a shortage of seats. The results can be reconciled by noting that in our model, if there is a shortage of seats and lists are long, the crossing point of the distributions occurs in the far right tail. As such, the findings can be viewed as complementary: our work shows that a tradeoff always exists, and they point out circumstances in which this tradeoff almost vanishes.

Theorem 3 identifies that the differences between lottery procedures may be significant. It is simultaneously a generalization of the main result in Ashlagi et al. (2019) (because it allows for heterogeneous popularity and list length), and a special case (because it addresses only first choices). Ashlagi et al. (2019) also show that under a single lottery, at least half of assigned students receive their first choice. Proposition 3 agrees with this finding when schools have a single seat and are equally popular. In addition, it shows how this bound decays as school popularity becomes more skewed, and how it improves as capacities grow.

In summary, our model allows us to study a new question (comparing match size across priority rules), and to identify the key role of a parameter that has been largely ignored by prior work (the distribution of list lengths). When specialized to the case of long lists sampled uniformly at random, it agrees with prior work that a single lottery produces better outcomes in markets with a shortage of seats. Furthermore, it shows how this finding generalizes to new settings: Theorem 2 points out that a tradeoff exists when list length is bounded, and Proposition 3 quantifies the decay in first choices when student interest concentrates on a small number of schools.

3. MODEL

We first introduce a model with ex-ante homogeneous schools. The model has four parameters:

- $\rho \in \mathbb{R}_+$, interpreted as the ratio of students to schools.
- $C \in \mathbb{N}$, interpreted as the capacity of each school.
- $\mathcal{L} : \mathbb{N} \rightarrow [0, 1]$, with $\mathcal{L}(k)$ interpreted as the fraction of students who list at most k schools.
- Priority rule $R \in \{S, I\}$.

$R = S$: *Single lottery*. Each student has a priority number drawn $U[0, 1]$.

$R = I$: *Independent lotteries*. Each student has a priority number at each school, drawn independently and $U[0, 1]$.

In this model, higher priority is better. Additionally, the schools on each student's list are selected uniformly at random.⁶ This is for expositional purposes: Section 3.5 extends the model to allow schools to differ in their popularity and capacity. All of our main results hold for this general model.

Given $\rho, C, \mathcal{L}$, for each $R \in \{S, I\}$ and any $k, \ell \in \mathbb{N}$ we will define the quantity $F_R(k|\ell)$, which represents the fraction of students matched to one of their top k choices, among students with lists of length ℓ , when using the student-proposing Deferred Acceptance algorithm with priorities generated according to rule R .

We express $F_R(k|\ell)$ in terms of a function $\Lambda_R : [0, 1] \rightarrow \mathbb{R}_+$. The value $\Lambda_R(t)$ represents the average number of students who are “interested” in a given school and have priority above t at that school (when priorities are generated by rule R). We say that a student is *interested* in a school if she is not admitted to any preferred school. Put another way, a student is interested in all schools to which she proposes over the course of the Deferred Acceptance algorithm. Note that although student lists are independent, student interest is not: a student's interest in her third choice depends on whether she gets into her first two choices, which depends on how many other students are interested in these schools. In order to define the outcomes $F_R(k|\ell)$, we seek a tractable and consistent description of the level of interest at each school.

We make two large-market assumptions, which allow us to characterize the joint distribution of interest across schools using the single function Λ_R . Fix a student who lists ℓ schools, and let $t_k \in [0, 1]$ denote her priority at her k^{th} choice school. Let N_k be the number of students who have higher priority at school k and are interested in this school. We assume the following.

I **Poisson**. Each N_k follows a Poisson distribution.

II **Independence**. The random variables $\{N_k\}_{k=1}^\ell$ are independent.

We motivate these assumptions in Section 3.1. Readers who are uninterested in these motivations can skip to Section 3.2, which defines several important functions. Section 3.3 uses these functions to define a consistency condition for Λ_S , as well as an expression for $F_S(k|\ell)$ in terms of Λ_S . Section 3.4 provides analogous derivations for Λ_I and $F_I(k|\ell)$. Section 3.5 describes the extension to heterogeneous schools and establishes uniqueness of consistent interest functions Λ_S and Λ_I .

3.1. Motivation.

Consider a model with a finite number of students and schools, where each school has some integral capacity. Each student has a preference list (an ordered subset of the set of schools). At each school on her list, she has a priority score, and no two students have the same score at any school. The Deferred Acceptance algorithm starts with each student expressing interest in the first school on her list (this is sometimes referred to as a “proposal” or an “application”). From there, the following steps are repeated until convergence:

- i. Each school rejects any students for which the number of higher-priority interested students is at least the school's capacity.

⁶Formally, we do not define a market with a continuum of students selecting lists independently, as this would require Kolmogorov's extension theorem and introduce unnecessary technicalities. We avoid this by defining outcomes as the solution to equations that depend on $\rho, C, \mathcal{L}$, and are inspired by thinking about a market with a continuum of students and schools.

- ii. Each student who has been rejected from every school at which she has expressed interest expresses interest in the next school on her list (if any such school exists).

When preferences and priorities are generated randomly, one way to study this algorithm is to generate them “on demand”: that is, waiting to reveal the k^{th} school on a student’s list (and her priority there) until it is known that she is interested in this school (Pittel (1992) and Immorlica and Mahdian (2005) refer to this as the “principle of deferred decisions”). In a model with iid uniform preferences, the next school is selected uniformly at random from the schools that the student has not already expressed interest in. If the number of schools is much larger than the number of schools on the student’s list, this is similar to selecting a school uniformly at random (with replacement). Thus one can approximate the amount of interest at any small set of schools using a “independent balls in urns” framework.

Fact 1. Fix $\ell \in \mathbb{N}$ and $\lambda \in \mathbb{R}_+$, and let $\{N_i\}_{i=1}^\ell$ be iid Poisson random variables with mean λ . If $\lfloor \lambda n \rfloor$ balls are thrown independently and uniformly at random into n urns, and B_i^n represents the number of balls in urn i , then $\{B_i^n\}_{i=1}^\ell$ converges in distribution to $\{N_i\}_{i=1}^\ell$ as $n \rightarrow \infty$.

In the school choice market, each ball represents a student expressing interest in a new school, and each urn corresponds to one school. This fact motivates our large market assumptions: interest at each school follows a Poisson distribution, and interest is independent across the schools on a student’s list.

3.2. Preliminaries. In this section, we define several useful functions derived from the Poisson distribution. Given $\lambda \in \mathbb{R}_+$, let

$$\begin{aligned}
 (1) \quad \mathcal{V}(\lambda) &= \sum_{k=0}^{C-1} \frac{e^{-\lambda} \lambda^k}{k!}, \\
 (2) \quad \mathcal{E}(\lambda) &= \sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} \min(k, C), \\
 (3) \quad \mathcal{A}(\lambda) &= \mathcal{E}(\lambda)/\lambda, \quad \mathcal{A}(0) = 1.
 \end{aligned}$$

These can be interpreted as follows. At a school where the number of interested students follows a Poisson distribution with mean λ , $\mathcal{V}(\lambda)$ gives the probability that the school will have at least one vacancy, $\mathcal{E}(\lambda)$ gives the average number of enrolled students, and $\mathcal{A}(\lambda)$ is its aggregate Acceptance rate. Given these interpretations, it is intuitive that

$$(4) \quad \mathcal{E}'(\lambda) = \mathcal{V}(\lambda),$$

meaning that if the expected number of interested students increases slightly, expected enrollment rises precisely by the probability that the school previously had a vacancy. A proof of (4) is in Lemma 2 in the Appendix.

We now define one more important function. Recall that a student expresses interest in the k^{th} school on her list if and only if she is rejected by the first $k - 1$. For a student who has list length drawn from $\mathcal{L}$ and is accepted independently with probability α at each listed school, the expected

number of schools in which she is interested is given by

$$(5) \quad \mu(\alpha) = \mathbb{E}_{\ell \sim \mathcal{L}} \left[\sum_{k=0}^{\ell-1} (1-\alpha)^k \right] = \sum_{k=0}^{\infty} (1 - \mathcal{L}(k)) (1-\alpha)^k.$$

Note that the second equality holds by writing the expectation as a sum and exchanging the order of summation. Note that $\mu(0) = \sum_{k \geq 0} (1 - \mathcal{L}(k))$ is the average list length, which we assume to be finite. Meanwhile, $\mu(1)$ is the fraction of students listing at least one school, which we assume without loss of generality to be one.

3.3. Single Lottery. To analyze outcomes for a student with priority t , suppose that we knew the average number of interested students with priority above t , $\Lambda_S(t)$. Assumption I states that the number of interested students with priority above t is Poisson. Therefore, the probability that this number is less than C (implying that the student with priority t is accepted) is $\mathcal{V}(\Lambda_S(t))$.

Students are always interested in their first school. They are interested in their k^{th} school only if their first $k-1$ schools reject them. Because the acceptance probability is $\mathcal{V}(\Lambda_S(t))$ and Assumption II states that outcomes are independent across schools, on average, a student with priority t is interested in $\mu(\mathcal{V}(\Lambda_S(t)))$ schools. This allows us to calculate Λ_S as the solution to the following differential equation:

$$(6) \quad \Lambda_S(1) = 0.$$

$$(7) \quad \Lambda'_S(t) = -\rho \mu(\mathcal{V}(\Lambda_S(t))).$$

Given Λ_S , we can calculate the fraction of students with list length ℓ who receive one of their top k choices. Each time a student with priority t is about to express interest in the next school on her list, her probability of being accepted by this school is $\mathcal{V}(\Lambda_S(t))$. It follows that the probability that she fails to receive one of her first k choices is $(1 - \mathcal{V}(\Lambda_S(t)))^{\min(k, \ell)}$. Integrating across priorities, we thus define

$$(8) \quad F_S(k|\ell) = 1 - \int_0^1 (1 - \mathcal{V}(\Lambda_S(t)))^{\min(k, \ell)} dt.$$

3.4. Independent Lotteries. Note that each time a student expresses interest in a school, her priority is drawn uniformly at random, independent of all other randomness. Therefore, the probability that the interested student has priority above t is $1-t$. That is,

$$(9) \quad \Lambda_I(t) = (1-t)\Lambda_I(0).$$

When a student expresses interest in a school at which her priority is t , Assumption I states that the number of higher-priority students that are interested in that school follows a Poisson distribution with mean $\Lambda_I(t)$, and therefore the probability of acceptance is $\mathcal{V}(\Lambda_I(t))$. Averaging over possible values of the priority t , each time a student expresses interest, she is accepted with probability

$$\int_0^1 \mathcal{V}(\Lambda_I(t)) dt = \int_0^1 \mathcal{V}((1-t)\Lambda_I(0)) dt = \mathcal{E}(\Lambda_I(0))/\Lambda_I(0) = \mathcal{A}(\Lambda_I(0)).$$

The first equality follows from (9), the second from (4), and the third by definition of $\mathcal{A}$ in (3). Because Assumption II states that levels of interest are independent across schools, the probability that a student is rejected from her first $k \leq \ell$ choices is $(1 - \mathcal{A}(\Lambda_I(0)))^k$, and the average student expresses interest in $\mu(\mathcal{A}(\Lambda_I(0)))$ schools. Of course, the scalar $\Lambda_I(0)$ must satisfy the following consistency condition:

$$(10) \quad \Lambda_I(0) = \rho \cdot \mu(\mathcal{A}(\Lambda_I(0))).$$

Given Λ_I , we can calculate the fraction of students with list length ℓ who receive one of their top k choices as follows:

$$(11) \quad F_I(k|\ell) = 1 - (1 - \mathcal{A}(\Lambda_I(0)))^{\min(k, \ell)}.$$

3.5. School Heterogeneity. We now extend the model to allow schools to differ in their capacity and popularity. There is a countable school type space $\mathcal{T} \subseteq \mathbb{N} \times \mathbb{R}_+$, and a distribution $\mathcal{D}$ on $\mathcal{T}$. For each type $\tau \in \mathcal{T}$, $\mathcal{D}_\tau \in [0, 1]$ represents the fraction of schools of type τ . Each $\tau \in \mathcal{T}$ is a pair (C_τ, p_τ) , where C_τ represents the capacity of schools of type τ , and p_τ represents “popularity” of these schools, as we now describe.

Instead of drawing schools uniformly at random, school types are sampled iid: for any $k \leq \ell$, the k^{th} school on a student’s list is of type τ with probability $\mathcal{D}_\tau p_\tau$. Hence, the average popularity $\sum_{\tau \in \mathcal{T}} \mathcal{D}_\tau p_\tau$ is one. We assume that the second moment of school popularity $\sum_{\tau \in \mathcal{T}} \mathcal{D}_\tau p_\tau^2$ is finite.

We maintain assumptions I and II, and generalize $\mathcal{V}(\cdot)$ and $\mathcal{E}(\cdot)$ as follows:

$$(12) \quad \mathcal{V}(\lambda) = \sum_{\tau \in \mathcal{T}} \mathcal{D}_\tau p_\tau \nu(p_\tau \lambda, C_\tau),$$

$$(13) \quad \mathcal{E}(\lambda) = \sum_{\tau \in \mathcal{T}} \mathcal{D}_\tau \varepsilon(p_\tau \lambda, C_\tau),$$

where ν and ε are our new names for the expressions in (1) and (2):

$$(14) \quad \nu(\lambda, C) = \sum_{k=0}^{C-1} \frac{e^{-\lambda} \lambda^k}{k!} = \mathbb{P}(\mathbf{Po}(\lambda) < C),$$

$$(15) \quad \varepsilon(\lambda, C) = \sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} \min(k, C) = \mathbb{E}[\min(\mathbf{Po}(\lambda), C)].$$

The basic model introduced previously corresponds to a singleton type space $\mathcal{T} = \{(C, 1)\}$, in which case the expressions in (12) and (13) coincide with (1) and (2). The motivation for these generalizations is as follows. If on average, schools receive interest from λ students, then the average interest at a school of type τ is $p_\tau \lambda$ (because the fraction $\mathcal{D}_\tau$ of schools with type τ receive a fraction $\mathcal{D}_\tau p_\tau$ of student interest).⁷ Therefore, the quantity $\mathcal{V}(\lambda)$ can be interpreted as the probability that the next time a student expresses interest, it will be at a school with a $\mathcal{V}$ acancy (given that on

⁷Returning to Fact 1, we note that it can be generalized as follows. If λn balls (representing student interest) are dropped into n urns (representing schools), with each ball falling in urn i with probability p_{τ_i}/n , then the number of balls in urn i is approximately Poisson with mean $p_{\tau_i} \lambda$, and is approximately independent from the number of balls in any finite collection of other urns.

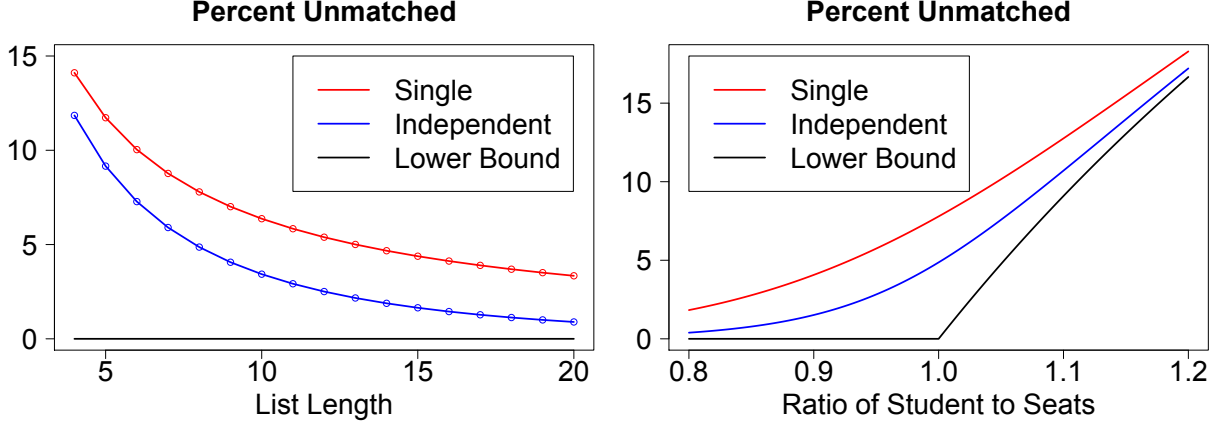


FIGURE 2. Comparison of the number of unmatched students when using single or independent lotteries. Schools are equally popular and have capacity $C = 1$, and all students list the same number of schools. On the left, the market is balanced ($\rho = 1$) and the list length ℓ varies. On the right, students list $\ell = 8$ schools, and the ratio of students to schools ρ varies. In these examples, independent lotteries consistently match more students. Corollary 1 states that this holds even with arbitrary heterogeneity among schools. Theorem 1 states that when students submit lists of different lengths, independent lotteries assign more students so long as $f(\alpha) = 1/\mu(\alpha)$ is convex, but this reverses if f is concave.

average, schools have interest from λ students). Similarly, $\mathcal{E}(\lambda)$ represents the average Enrollment across schools.

Our main results express match outcomes in terms of the functions Λ_R . The following proposition establishes that these functions are well-defined: that is, that there are unique consistent functions Λ_S and Λ_I .

Proposition 1. *Let $\mathcal{V}(\cdot)$ and $\mathcal{E}(\cdot)$ be as given in (12), (13), let $\mathcal{A}(\cdot)$ be as given in (3), and let μ be as given in (5). For any $\rho, \mathcal{T}, \mathcal{D}, \mathcal{L}$, there exists a unique function $\Lambda_S : [0, 1] \rightarrow \mathbb{R}_+$ satisfying (6) and (7), and a unique function $\Lambda_I : [0, 1] \rightarrow \mathbb{R}_+$ satisfying (9) and (10).*

4. RESULTS

We will study properties of the statistics $F_R(k|\ell)$, which represent the fraction of students listing ℓ schools who receive one of their first k choices under priority rule R . The number of students who receive one of their first k choices is given by the weighted average

$$(16) \quad F_R(k) = \mathbb{E}_{\ell \sim \mathcal{L}} [F_R(k|\ell)].$$

Note that $F_R(\infty)$ represents the fraction of students who match to some school on their list.

4.1. Number of Assigned Students.

We first compare the percentage of students that are left unassigned under each lottery procedure. Figure 2 displays these percentages for one-to-one markets with homogeneous schools, as student list length ℓ and the ratio of students to schools ρ vary. For these parameter values, independent

lotteries appear always to match more students. Does this continue to hold even if schools differ in their popularity and capacity? What if students submit lists of different lengths?

Corollary 1 establishes that the answer to the first question is ‘yes’: if students all list the same number of schools, then independent lotteries result in more matches, regardless of the number of students, the length of their lists, or the size and popularity of different schools. The answer to the second question is more nuanced. Theorem 1 establishes that the comparison of unassigned students hinges on the shape of the list length distribution (and no other market parameters).

Theorem 1. *Fix any $\rho, \mathcal{T}, \mathcal{D}$. If the list length distribution $\mathcal{L}$ is such that $f(\alpha) = 1/\mu(\alpha)$ is*

- *concave, then more students are assigned under a single lottery: $F_I(\infty) \leq F_S(\infty)$.*
- *convex, then more students are assigned under independent lotteries: $F_I(\infty) \geq F_S(\infty)$.*

This provides a clean characterization of when independent lotteries result in more matches than a single lottery. This holds if the function f – which depends on the list length distribution $\mathcal{L}$ but no other primitives – is convex. Conversely, if f is concave, a single lottery results in more matches. If f is linear, the two procedures lead to the same number of matches.

One might wonder which case (if either) should be “typical.” Figure 3 displays list length data from New York City, Amsterdam, Hungary, and Chile. In all four cases the function f is convex, so Theorem 1 suggests that using a single lottery to break ties should result in fewer assigned students. This prediction is consistent with the simulation results presented by Abdulkadiroglu et al. (2009) and de Haan et al. (2018).

Proposition 2 provides simple sufficient conditions for concavity and convexity of f based on whether the list length distribution $\mathcal{L}$ has an increasing or decreasing hazard rate.

Definition 1. *Define the **hazard rate** of $\mathcal{L}$ at k to be $\mathbb{P}_{\ell \sim \mathcal{L}}(\ell = k | \ell \geq k) = \frac{\mathcal{L}(k) - \mathcal{L}(k-1)}{1 - \mathcal{L}(k-1)}$.*

*The distribution $\mathcal{L}$ is **light-tailed** if its hazard rate is weakly increasing in k , and **heavy-tailed** if its hazard rate is weakly decreasing in k .*

Proposition 2.

- I. *If the list length distribution $\mathcal{L}$ is heavy-tailed, then $f(\alpha) = 1/\mu(\alpha)$ is concave.*
- II. *If the list length distribution $\mathcal{L}$ is light-tailed, then $f(\alpha) = 1/\mu(\alpha)$ is convex.*

A deterministic distribution is light-tailed, so Theorem 1 and Proposition 2 imply the following.

Corollary 1. *If all students list the same number of schools, then independent lotteries match more students than a single lottery.*

Other examples of light-tailed distributions include uniform distributions, Poisson distributions, and truncated Poisson distributions. An example of a heavy-tailed distribution is $\mathcal{L}(k) = 1 - (k + 1)^{-2}$. The geometric distribution has a constant hazard rate, and is both light-tailed and heavy-tailed. Any heavy-tailed distribution is necessarily unbounded, which is permitted by our model but could not arise in practice. Although there exist distributions with bounded support for which f is concave, the convex case appears to be much more common.

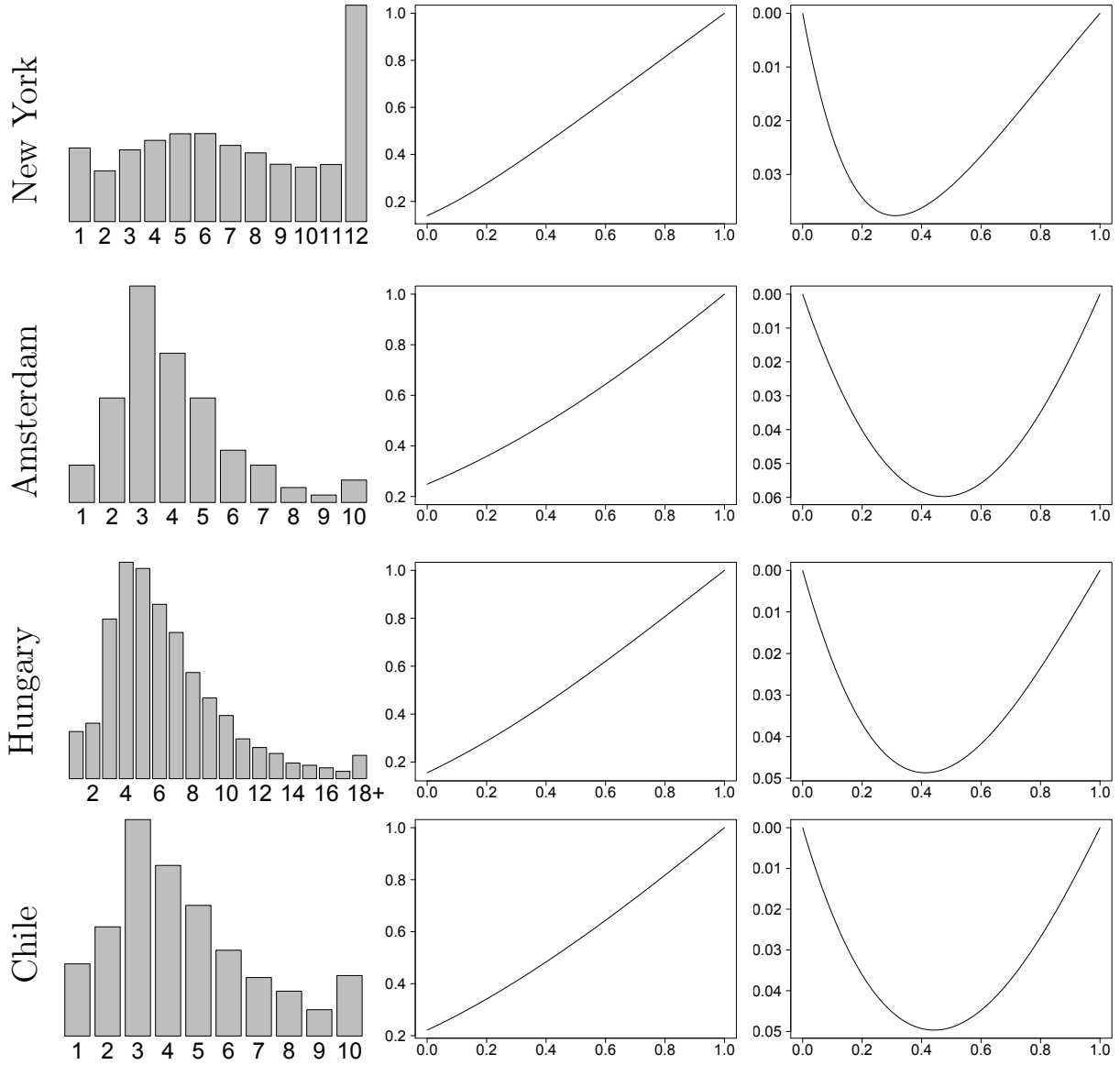


FIGURE 3. Left panel: list length distributions from four centralized admissions programs. Middle panel: the corresponding functions $f(\alpha)$. Despite differences in the average list length and the shape of the distributions, in all cases, f is convex, as seen by removing the linear trend and plotting $f(\alpha) - \alpha f(1) - (1 - \alpha)f(0)$ (right panel). Theorem 1 suggests that when f is convex, breaking ties using a single lottery should match fewer students. This prediction is consistent with the empirical findings from New York (Abdulkadiroglu et al. (2009)) and Amsterdam (de Haan et al. (2018)). Hungarian list length data taken from Aue et al. (2019), and Chilean data from Larroucau and Rios (2019).

4.2. Student Rank Distributions. We next study the full rank distribution. Figure 4 plots the fraction of students who receive one of their first k choices under each procedure, for a particular example. In this example, students all list the same number of schools, so Corollary 1 establishes that independent lotteries result in more matches (that is, the blue line is higher at $k = 10$). We

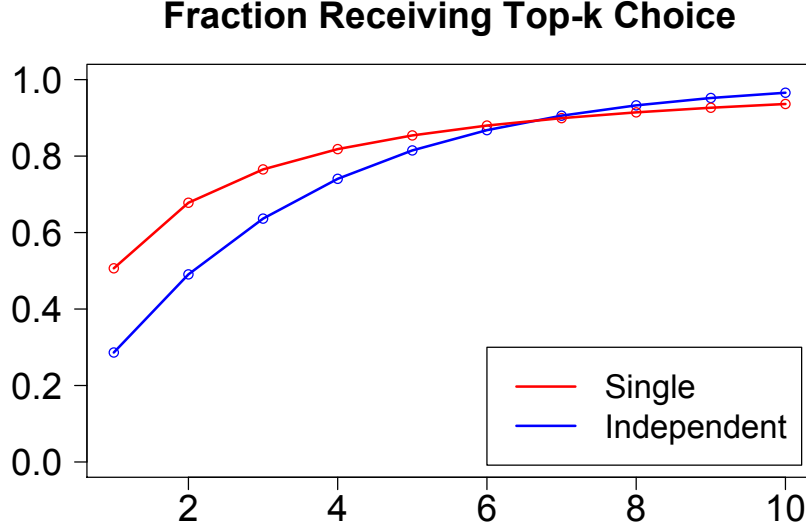


FIGURE 4. The fraction of students assigned to a top- k choice among students who list at least k schools. In this example, schools are equally popular and have a single seat, the market is balanced, and all students list $\ell = 10$ schools. Independent lotteries assign fewer students to their top choice, but more students overall. Furthermore, the lines cross exactly once. Theorem 2 establishes that these observations hold with arbitrary school heterogeneity, market imbalance, and list length.

observe in Figure 4 that a single lottery results in more students receiving their first choice, as well as more students receiving one of their top k choices for $k \leq 6$. However, for $k \geq 7$, more students get one of their first k choices under independent lotteries. The following result establishes that this “single crossing” property holds generally.

Theorem 2. *Given any $\rho, \mathcal{L}, \mathcal{T}, \mathcal{D}$, there exists $k' \in \mathbb{N}$ with $\mathcal{L}(k') < 1$ such that:*

- I. *If $k \leq k'$, then $F_I(k|\infty) < F_S(k|\infty)$.*
- II. *If $k > k'$, then $F_I(k|\infty) \geq F_S(k|\infty)$.*

Note that the quantity $F_R(k|\infty)$ represents the fraction of students who receive one of their first k choices *among those who list at least k schools*.⁸ If all student lists have the same length, then Theorem 2 states that the weighted averages $F_I(k)$ and $F_S(k)$ defined by (16) cross once, but in general this need not be the case.⁹ However, Theorem 2 does imply that for “small” k , more students receive one of their top- k choices when using a single lottery. In particular, a single lottery always gives more students their first choice.

Corollary 2. *For any $\rho, \mathcal{L}, \mathcal{T}, \mathcal{D}$, more students receive their first choice under a single lottery than under independent lotteries: $F_I(1) \leq F_S(1)$.*

⁸The definitions in (8) and (11) imply that $F_R(k|\ell)$ depends only on the *minimum* of k and ℓ : among students listing $k, k+1, k+2, \dots$ schools, the same fraction receive one of their top k choices. Thus, Theorem 2 could equivalently use $F_R(k|k)$ instead of $F_R(k|\infty)$.

⁹In fact, it is possible that F_I and F_S cross multiple times, or not at all. For examples, let $\mathcal{T}$ be the singleton $(1, 1)$ (so schools are equally popular and have a single seat), and let $\rho = 1$. If students are equally likely to list 2 or 12 schools, F_I and F_S cross twice: a single lottery assigns more students to their first choice and more students overall, but assigns fewer students to one of their top four choices. If 80% of students list a single school, and the remaining 20% list three schools, then F_I and F_S never cross.

If we call a list *short* if $\ell \leq k'$, and *long* if $\ell > k'$, then Theorem 2 implies the following:

- For students submitting short lists, the rank distribution under a single lottery dominates the rank distribution under independent lotteries.
- For students submitting long lists, more match under independent lotteries, but more get a top- k choice (for $k \leq k'$) under a single lottery.

That is, students with short list unambiguously prefer a single lottery, whereas those with long lists face a tradeoff.

4.3. Magnitudes of Differences. Although Corollary 2 states that more students are assigned to their first choice when using a single lottery, it is silent on the magnitude of the difference. Of course, if the number of students (parameterized by ρ) is very large, then the fraction of students who get their first choice is necessarily small under any allocation procedure. Accordingly, it makes sense to study the fraction of *assigned* students who receive their first choice. Theorem 3 provides a tight lower bound on this quantity when using each procedure.

Theorem 3. *Fix $\mathcal{T}, \mathcal{D}$. Under a single lottery, the fraction of assigned students who receive their first choice is at least*

$$\inf_{\rho, \mathcal{L}} \frac{F_S(1)}{F_S(\infty)} = \frac{\int_0^\infty \mathcal{V}(\lambda)^2 d\lambda}{\int_0^\infty \mathcal{V}(\lambda) d\lambda},$$

whereas under independent lotteries,

$$\inf_{\rho, \mathcal{L}} \frac{F_I(1)}{F_I(\infty)} = 0.$$

The fraction of assigned students who receive their first choice under a single lottery is lower-bounded by a quantity that depends on school capacities and popularity, but is independent of the number of students and the list length distribution.¹⁰ By contrast, under independent lotteries, if students outnumber seats, the number of students receiving their first choice goes to zero as student lists grow. Therefore, when students submit long lists, the number of students who receive their first choice can be quite different under the two procedures.

The lower bound from Theorem 3 can be explained as follows. The worst case occurs when students outnumber seats, and all students submit long lists. In this case, all schools will receive a lot of interest, and all seats will fill. Recalling that $\mathcal{V}(\lambda) = \mathcal{E}'(\lambda)$ (see (4)), the denominator can be re-expressed as $\mathcal{E}(\infty) = \sum_\tau \mathcal{D}_\tau C_\tau$. That is, it represents the total number of matches. Meanwhile, for students with priority t , a fraction $\mathcal{V}(\Lambda(t))$ of their proposals are accepted. Therefore, they express interest in an average of $1/\mathcal{V}(\Lambda(t))$ schools for each acceptance, implying that only a fraction $\mathcal{V}(\Lambda(t))$ of their proposals go to their first choice. It follows that among these students, the fraction of all proposals that both go to a first choice school and are accepted is $\mathcal{V}(\Lambda(t))^2$. Adding this up across students reveals that the number of first-choices is lower-bounded by the numerator.

We now discuss properties of this bound. Unsurprisingly, the bound is largest when schools are equally popular, and close to zero if schools differ dramatically in their popularity. The following

¹⁰With uniform preferences and one-to-one matching, the lower bound for a single lottery becomes 1/2. As mentioned in Section 2, this result appears in Ashlagi et al. (2019). Theorem 3 extends this result to give tight bounds for arbitrary distributions of school popularity and capacity.

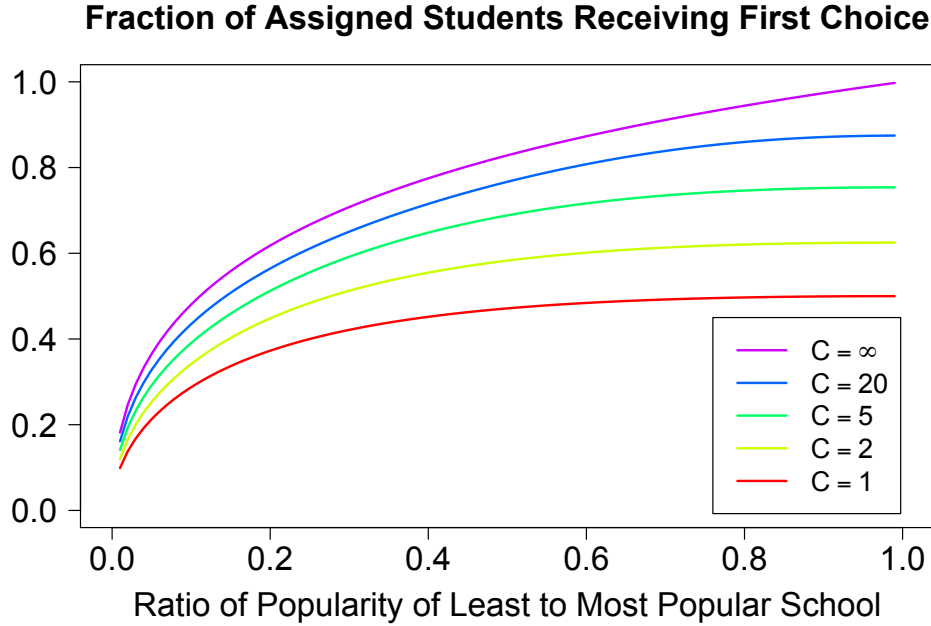


FIGURE 5. Using a single lottery ensures that a sizeable fraction of assigned students receive their first choice school. The figure shows tight lower bounds on this quantity when schools have identical capacity C . When schools are equally popular, at least half of assigned students receive their first choice, and this number rises as C grows. As the discrepancy in popularity rises, the number of students receiving their first choice falls. Proposition 3 states that the bound for $C = 1$ is $\sqrt{r}/(1+r)$; as $C \rightarrow \infty$ it approaches $2(\sqrt{r}-1)/(r-1)$.

result quantifies the decay in the number of students receiving their first choice as discrepancies in popularity become more pronounced.

Proposition 3. *If $C_\tau = 1$ for all $\tau \in \mathcal{T}$ and the most popular school is at most r times as popular as the least popular school, then*

$$\frac{F_S(1)}{F_S(\infty)} \geq \frac{\sqrt{r}}{1+r}.$$

This implies that if no school is more than four times more popular than any other, then at least 40% of assigned students get their top choice. If no school is more than nine times more popular than any other, then at least 30% of assigned students get their top choice. Appendix A.3 derives analogous bounds when $C_\tau = C > 1$ for all τ .¹¹ As the capacity C increases, the lower bound increases, as shown in Figure 5. As $C \rightarrow \infty$, the bound approaches $2\frac{\sqrt{r}-1}{r-1}$, implying that at least $2/3$ of assigned students receiving their first choice when $r = 4$ and $1/2$ of assigned students receiving their first choice when $r = 9$.

¹¹The proof proceeds as follows. First, we show that the worst case distribution features only two types of school: popular and unpopular. Next, we show that the worst case is for the fraction of popular schools to be $(\sqrt{r}-1)/(r-1)$. Finally, we evaluate the number of first choices under this worst case distribution. For $C = 2$, the resulting lower bound is $\left(\frac{\sqrt{r}-1}{r-1}\right)^2 \left(\frac{5}{4}\sqrt{r} + \frac{2r}{r+1} + \frac{2r^2}{(r+1)^3}\right)$.

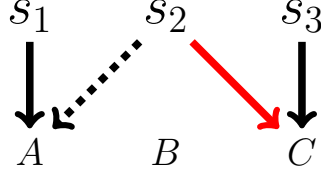


FIGURE 6. A small example providing intuition for the results of Sections 4.1 and 4.2. Each school has a single seat, and students' first choices are in black. School A rejected s_2 , and her second choice is C . Under independent lotteries, C is equally likely to reject s_2 and s_3 , whereas under a single lottery, s_2 is more likely to be rejected. If s_2 is rejected, both s_1 and s_3 will get their first choice. The list length distribution determines whether s_2 or s_3 faces greater risk of going unassigned.

We could also study the difference in the number of matches. In the examples in Figure 2, the absolute difference is about 3%. For example, independent lotteries with $\ell = 10$ leave 3.4% of students unassigned, compared to 6.4% under a single lottery. If there are more seats than students, then under both procedures, the number of unassigned students goes to zero as student list length grows. However, lists may have to be very long for this convergence to kick in: in the aforementioned example, to reduce the number of unassigned students below 3.4% while using a single lottery, lists must be lengthened to $\ell = 20$ schools (at which point independent lotteries leave only 0.9% of students unmatched). In Appendix A.4, we show that in balanced one-to-one markets with homogeneous schools, the number of unassigned students decays exponentially in the square root of the list length ℓ under independent lotteries, whereas it decays at rate $1/\ell$ under a single lottery. When there are fewer students than seats, the number of unassigned students decays exponentially in ℓ under both procedures, but at very different rates.

5. INTUITION VIA EXAMPLE

The intuition for many of our results can be illustrated through the simple example depicted in Figure 6. In this example, there are three students and three schools, each with a single seat. Students s_1 and s_2 both listed school A as their first choice. Student s_2 was rejected, and listed C second. If C rejects s_2 , then both s_1 and s_3 will get their first choice; if C rejects s_3 , at most one student will receive her first choice. Under independent lotteries, C is equally likely to reject each student, whereas under a single lottery, the fact that s_2 has previously been rejected suggests that she has a poor lottery number, and is likely to be rejected again. Therefore, the expected number of students receiving their first choice is higher under a single lottery. This reasoning, which is similar to that presented by Ashlagi and Nikzad (2019) and Ashlagi et al. (2019), applies whenever a student is competing for her top choice with another student who has already been rejected several times.

We can also use this example to illustrate why the comparison of the number of matches depends on the shape of the list length distribution. Let us first suppose that every student lists exactly two schools. In that case, s_2 will go unassigned if rejected by C , whereas s_3 might match elsewhere. Therefore, the expected number of matches is higher when s_3 is rejected, which (as discussed above) is more likely under independent lotteries. This conclusion reverses if we assume that most students

are selective and list only one school, while a few students list all three. In that case, we know that s_2 is not selective: if rejected from C , she will be assigned to B . Meanwhile, it is likely that s_3 listed only school C , and will go unassigned if she does not receive it. Therefore, the expected number of matches is highest when s_2 is rejected, which is more likely under a single lottery. This reasoning suggests that so long as students who have been rejected more often tend to have fewer *remaining* options than those that have been rejected less often, then independent lotteries will lead to more matches. Otherwise, a single lottery will. The hazard rate condition in Proposition 2 makes this intuition precise.

Although our model addresses only single and independent lotteries, this intuition extends to other priority rules. So long as students who have been rejected more often tend to have fewer options remaining on their list, there will be a tradeoff between assigning students to their top choice and assigning as many students as possible. As one example, Arnosti (2019) considers a setting where each (student, school) pair has an iid match quality, and both priorities and preferences are generated by ranking according to this quality. While this intuitively seems like it should lead to “good” outcomes (after all, each student’s favorite school is likely to rank her highly!), those that fail to get their favorite school may face a sequence of rejections. In fact, that paper shows that the expected match size is *lower* than would result from using a single lottery at each school (and thus, by Theorem 1, also lower than using independent lotteries if $1/\mu(\alpha)$ is convex).

In practice, we can apply this thinking to the widespread policy of giving priority to students who already have a sibling at the school. The reasoning above suggests that sibling priority likely increases the number of students who are assigned to their first choice. However, because it makes it harder for students without sibling priority to be admitted, it might result in fewer matches overall. If minimizing the number of unassigned students is important, then *negatively* correlated school lotteries might be appropriate.

6. DISCUSSION

This work addresses an operational question facing school district administrators in cities across the world: how to implement lotteries that generate priority lists? Our findings suggest that there is typically a tradeoff between maximizing the number of students who get their first choice, and matching as many students as possible. Therefore, the appropriate choice likely depends on the value that the district attaches to each of these objectives. Of course, the magnitude of differences between procedures is also important. In practice, it appears that a single lottery often matches significantly more students to their first choice, at little cost in match size. Although comparing magnitudes is not our main focus, these findings are in line with the predictions from Section 4.3 and Appendix A.4.

Although we focus on lottery design, a second contribution of this work is a continuum model that could be used to answer many other questions pertaining to school choice. Its usefulness stems from the ability to incorporate elements that have proved difficult to analyze in finite markets: schools differ arbitrarily in their popularity and capacity, and the length of student lists follows an arbitrary distribution. As a result, our finding that the shape of the list length distribution plays a key role in the comparison of match size has no parallel in the literature.

For one example of a question that merits further study, note that policymakers often encourage students to list more schools, in order to reduce the change of going unassigned. It is known that adding a school to the bottom of a student’s list (weakly) increases the number of matches, while (weakly) worsening outcomes for all other students, but there is no guidance on when these effects are “large.” It is possible that encouraging students to submit longer lists actually results in worse outcomes for most students! Our model is well-suited to attacking this question.¹²

Related to the topic of influencing student lists, we should point out that our comparisons hold for fixed lists submitted by students. One natural question is whether changes in priority might induce changes in submitted preferences. In theory, this should not be a concern: when using student-proposing Deferred Acceptance, it is a dominant strategy for students to submit their true preferences, regardless of how priorities are determined (Roth, 1984).

In practice, some students would almost certainly respond to changes in priority. For example, a student who is guaranteed admission to her neighborhood school may not bother to list schools that are inferior to this option, but would list these schools if neighborhood priority were removed. Students who are initially uncertain about where they would like to attend may focus their research on schools where they expect to be admitted. Furthermore, most school districts limit the number of schools that each student can rank (Pathak and Sönmez, 2013), implying that even students who know their own preferences might strategically list schools where they have high priority.

Despite these concerns, we think that the assumption that lists do not change is a reasonable point to start analysis. Regarding list length constraints, most students do not submit the maximum number of allowed schools. For example, this number is approximately 22% in New York, 27% in Denver, and 5% in New Orleans (Pathak, 2016). Additionally, even for students who submit the maximum number of schools, it is not clear that their reports respond significantly to the mechanism. As evidence, consider a policy change made by Chicago’s exam schools in 2009. Before the change, these schools had been assigned using a variant of the Boston mechanism. Preferences were collected from over 13,000 families in 2009, before a decision was made to dramatically change the mechanism (to Deferred Acceptance with a list length limit) and the priorities (giving affirmative action based on socioeconomic status rather than race). Students were allowed to resubmit their preferences, but only 11% made any changes (Pathak, 2016). Pathak (2016) concludes, “The magnitude of the behavioral response is swamped, for instance, by the mechanical change in how applicants are processed by the mechanism.” Meanwhile, Hastings et al. (2009) study data from Charlotte, North Carolina, and conclude that there is little evidence that applicants’ preferences reacted to a change in neighborhood boundaries. Thus, while changes to the assignment mechanism no doubt affect some students’ lists, the effect appears to be modest even for dramatic changes, and would likely be smaller for changes such as the implementation of the lottery.¹³

¹²The results in Section 4.3 and Appendix A.4 indicate that under a single lottery, lengthening lists imposes a limited externality on other students, but also may not be especially effective at reducing the number of unassigned students. Meanwhile, under an independent lottery, the externalities from long lists are much more severe. Future studies might investigate how these magnitudes depend on the distribution of school popularity, or the number of excess seats.

¹³In fact, Abdulkadiroglu et al. (2009) report that in the first year of implementation, policymakers in New York did not commit to which lottery procedure they would use, leaving students no opportunity to respond to this choice.

Conversely, for those who insist on taking seriously the idea that students generate lists strategically, we note that any analysis of this behavior requires a prediction for what options are available to each student (given the strategies employed by others). This has generally been viewed as intractable unless the students have perfect information or view all schools symmetrically. The model presented in this paper generates tractable predictions of an asymmetric market where students have incomplete information, and therefore could be used as a building block for the analysis of strategic listing.

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A. PROOFS

We begin with several technical lemmas that will come in handy. When possible, we include remarks interpreting the quantities involved.

Lemma 1. *Let N be a random variable on $\mathbb{N}$. Then $\mathbb{E}[N] = \sum_{k=0}^{\infty} \mathbb{P}(N > k)$.*

Proof.

$$\mathbb{E}[N] = \sum_{j=0}^{\infty} j \mathbb{P}(N = j) = \sum_{j=0}^{\infty} \sum_{k=0}^{j-1} \mathbb{P}(N = j) = \sum_{k=0}^{\infty} \sum_{j=k+1}^{\infty} \mathbb{P}(N = j) = \sum_{k=0}^{\infty} \mathbb{P}(N > k).$$

□

Lemma 2. *For $C \in \mathbb{N}$,*

$$\frac{d}{d\lambda} \nu(\lambda, C) = -\frac{e^{-\lambda} \lambda^{C-1}}{(C-1)!}, \quad \frac{d}{d\lambda} \varepsilon(\lambda, C) = \nu(\lambda, C), \quad \lim_{\lambda \rightarrow \infty} \varepsilon(\lambda, C) = C.$$

Therefore for any $\mathcal{T}, \mathcal{D}$,

$$|\mathcal{V}'(\lambda)| \leq \sum_{\tau \in \mathcal{T}} \mathcal{D}_{\tau} p_{\tau}^2, \quad \mathcal{E}'(\lambda) = \mathcal{V}(\lambda), \quad \lim_{\lambda \rightarrow \infty} \mathcal{E}(\lambda) = \sum_{\tau \in \mathcal{T}} \mathcal{D}_{\tau} C_{\tau}.$$

Proof. By (14),

$$(17) \quad \frac{d}{d\lambda} \nu(\lambda, C) = -\sum_{k=0}^{C-1} \frac{e^{-\lambda} \lambda^k}{k!} + \sum_{k=0}^{C-2} \frac{e^{-\lambda} \lambda^k}{k!} = -\frac{e^{-\lambda} \lambda^{C-1}}{(C-1)!}$$

By Lemma 1 it follows that

$$\varepsilon(\lambda, C) = \mathbb{E}[\min(\mathbf{Po}(\lambda), C)] = \sum_{k=0}^{C-1} \mathbb{P}(\mathbf{Po}(\lambda) > k) = \sum_{k=0}^{C-1} (1 - \nu(\lambda, k+1))$$

Therefore, by (17),

$$(18) \quad \frac{d}{d\lambda} \varepsilon(\lambda, C) = \frac{d}{d\lambda} \sum_{k=0}^{C-1} (1 - \nu(\lambda, k+1)) = \sum_{k=0}^{C-1} \frac{e^{-\lambda} \lambda^k}{k!} = \nu(\lambda, C).$$

From (14) and (17) it follows that

$$|\mathcal{V}'(\lambda)| = \left| \frac{d}{d\lambda} \sum_{\tau \in \mathcal{T}} \mathcal{D}_{\tau} p_{\tau} \nu(p_{\tau} \lambda, C_{\tau}) \right| = \sum_{\tau \in \mathcal{T}} \mathcal{D}_{\tau} p_{\tau}^2 \frac{e^{-p_{\tau} \lambda} (p_{\tau} \lambda)^{C_{\tau}-1}}{(C_{\tau}-1)!} \leq \sum_{\tau \in \mathcal{T}} \mathcal{D}_{\tau} p_{\tau}^2.$$

Similarly, (15) and (18) imply that $\mathcal{E}'(\lambda) = \mathcal{V}(\lambda)$.

Finally, as $\lambda \rightarrow \infty$, it is clear that $\varepsilon(\lambda, C) = \mathbb{E}[\min(\mathbf{Po}(\lambda), C)] \rightarrow C$, and $\mathcal{E}(\lambda) \rightarrow \sum_{\tau \in \mathcal{T}} \mathcal{D}_{\tau} C_{\tau}$.

□

Remark 1. *Lemma 2 has an intuitive explanation. The quantity $\varepsilon(\lambda, C)$ represents the expected number of seats filled at a school with C seats where an average of λ students express interest. Meanwhile, $\nu(\lambda, C)$ represents the probability that this school has at least one vacancy. Thus, the*

result $\frac{d}{d\lambda}\nu(\lambda, C) = -\frac{e^{-\lambda}\lambda^{C-1}}{(C-1)!}$ simply states that an additional student expressing interest eliminates the last vacancy if and only if $C - 1$ students had previously expressed interest. The result $\frac{d}{d\lambda}\varepsilon(\lambda, C) = \nu(\lambda, C)$ states that the increase in enrollment due to an additional student expressing interest is equal to the probability that the school previously had a vacancy. The result $\lim_{\lambda \rightarrow \infty} \varepsilon(\lambda, C) = C$ states that if the school has a lot of interest, it will almost always fill all seats.

Lemma 3. $\lambda\mathcal{A}'(\lambda) = \mathcal{V}(\lambda) - \mathcal{A}(\lambda) \leq 0$, with strict inequality for $\lambda > 0$.

Proof. The equality follows because

$$\mathcal{A}'(\lambda) = \frac{d}{d\lambda} \frac{\mathcal{E}(\lambda)}{\lambda} = \frac{\lambda\mathcal{V}(\lambda) - \mathcal{E}(\lambda)}{\lambda^2} = \frac{\mathcal{V}(\lambda) - \mathcal{A}(\lambda)}{\lambda}.$$

The inequality follows because

$$\mathcal{A}(\lambda) = \mathcal{E}(\lambda)/\lambda = \int_0^1 \mathcal{V}((1-t)\lambda)dt > \int_0^1 \mathcal{V}(\lambda)dt = \mathcal{V}(\lambda).$$

(Note that the first equality holds by definition, the second follows by the substitution $u = (1-t)\lambda$ and Lemma 2, and the inequality follows because $\mathcal{V}(\cdot)$ is decreasing.) $\square$

Remark 2. The quantity $\mathcal{V}(\lambda)$ represents the fraction of interest that is directed to schools with a vacancy. The quantity $\mathcal{A}(\lambda)$ represents the aggregate acceptance rate. Because schools with vacancies accept all interested students, it follows that $\mathcal{V}(\lambda) \leq \mathcal{A}(\lambda)$, and that their difference can be interpreted as the fraction of interest that triggers rejection of another student. Therefore, Lemma 3 states that the change in acceptance rate due to additional students expressing interest is equal to the number of additional rejections divided by the total amount of interest λ .

Lemma 4. For any $\mathcal{L}$, $\mu(\cdot)$ is decreasing and convex, with

$$(19) \quad \mu(\alpha) \leq \sum_{k=0}^{\infty} (1 - \mathcal{L}(k)) = \mathbb{E}_{\ell \sim \mathcal{L}}[\ell]$$

$$(20) \quad \mu'(\alpha) \leq \sum_{k=0}^{\infty} k(1 - \mathcal{L}(k)) = \frac{1}{2} \mathbb{E}_{\ell \sim \mathcal{L}}[\ell^2 - \ell].$$

Furthermore, $\alpha \cdot \mu(\alpha)$ is increasing in α , and at most 1.

Proof. The fact that $\mu(\alpha)$ is less than the average list length $\mu(0) = \sum_{k=0}^{\infty} (1 - \mathcal{L}(k))$ follows by inspection of (5). Differentiate (5) to see that

$$(21) \quad \mu'(\alpha) = -\sum_{k=0}^{\infty} k(1 - \mathcal{L}(k))(1 - \alpha)^k,$$

from which it follows that μ is decreasing and convex. Furthermore, (21) implies (20).

Monotonicity of $\alpha \cdot \mu(\alpha)$ holds because

$$\begin{aligned} \alpha \cdot \mu(\alpha) &= \sum_{k=0}^{\infty} (1-\alpha)^k (1 - \mathcal{L}(k)) - \sum_{k=0}^{\infty} (1-\alpha)^{k+1} (1 - \mathcal{L}(k)) \\ &= \sum_{k=0}^{\infty} (1-\alpha)^k (1 - \mathcal{L}(k)) - \sum_{k=1}^{\infty} (1-\alpha)^k (1 - \mathcal{L}(k-1)) \\ &= 1 - \sum_{k=1}^{\infty} (\mathcal{L}(k) - \mathcal{L}(k-1)) (1-\alpha)^k, \end{aligned}$$

which is clearly increasing in α . The claim that $\alpha \cdot \mu(\alpha) \leq 1$ follows by taking $\alpha = 1$. $\square$

Remark 3. *The intuition for Lemma 4 is as follows. Consider a student for whom each application is accepted with probability α . The quantity $\mu(\alpha)$ represents the expected number of schools at which this student will express interest. This quantity increases as the acceptance probability decreases, and is bounded by the average list length. The quantity $\alpha \mu(\alpha)$ represents the probability that this student matches, which is clearly less than one, and increases with α .*

Lemma 5. *For $R \in \{I, C\}$ the following equalities hold:*

$$(22) \quad \Lambda_R(0) = \rho \sum_{\ell=0}^{\infty} (1 - \mathcal{L}(\ell)) (1 - F_R(\ell|\infty)).$$

$$(23) \quad \mathcal{E}(\Lambda_R(0)) = \rho F_R(\infty).$$

Proof. We first prove (22) for $R = I$. This follows immediately from the definitions of Λ_I in (10), μ in (5), and F_I in (11):

$$\begin{aligned} \Lambda_I(0) &= \rho \mu(\mathcal{A}(\Lambda_I(0))) \\ &= \rho \sum_{k=0}^{\infty} (1 - \mathcal{L}(k)) (1 - \mathcal{A}(\Lambda_I(0)))^k \\ &= \rho \sum_{k=0}^{\infty} (1 - \mathcal{L}(k)) (1 - F_I(k|\infty)). \end{aligned}$$

We now prove (22) for $R = S$. By the definition of F_S in (8),

$$\begin{aligned} \rho \sum_{k=0}^{\infty} (1 - \mathcal{L}(k)) (1 - F_S(k|\infty)) &= \rho \sum_{k=0}^{\infty} (1 - \mathcal{L}(k)) \int_0^1 (1 - \mathcal{V}(\Lambda_S(t)))^k dt \\ &= \rho \int_0^1 \mu(\mathcal{V}(\Lambda_S(t))) dt \\ &= \int_0^{\Lambda_S(0)} d\lambda_S = \Lambda_S(0). \end{aligned}$$

The final line follows from the substitution $\lambda_S = \Lambda_S(t)$ and (7), which states that $\frac{d\lambda_S}{dt} = -\rho\mu(\lambda_S)$.

Turning to (23), we start with $R = I$, and let $\lambda_I = \Lambda_I(0)$. By definition of $\mathcal{A}$ in (3) we have

$$\begin{aligned}\mathcal{E}(\lambda_I) &= \lambda_I \mathcal{A}(\lambda_I) \\ &= \rho \mathcal{A}(\lambda_I) \mu(\mathcal{A}(\lambda_I)) \\ &= \rho (1 - \mathbb{E}_{\ell \sim \mathcal{L}} [(1 - \mathcal{A}(\lambda_I))^\ell]) \\ &= \rho F_I(\infty).\end{aligned}$$

The second equality follows from (10). The third follows from the definition of μ in (5) and the observation that

$$(24) \quad \alpha \mu(\alpha) = \alpha \mathbb{E}_{\ell \sim \mathcal{L}} \left[\sum_{k=0}^{\ell-1} (1 - \alpha)^k \right] = \mathbb{E}_{\ell \sim \mathcal{L}} [1 - (1 - \alpha)^\ell].$$

The fourth equality follows from the definitions of F_I in (11) and (16).

Finally, we prove (23) for $R = S$. By definition of F_S in (8) and (16), we have

$$\begin{aligned}F_S(\infty) &= \int_0^1 \mathbb{E}_{\ell \sim \mathcal{L}} [1 - (1 - \mathcal{V}(\Lambda_S(t)))^\ell] dt \\ &= \int_0^1 \mathcal{V}(\Lambda_S(t)) \mu(\mathcal{V}(\Lambda_S(t))) dt. \\ &= \frac{1}{\rho} \int_0^{\Lambda_S(0)} \mathcal{V}(\lambda_s) d\lambda_s \\ &= \mathcal{E}(\Lambda_S(0)) / \rho.\end{aligned}$$

The second equality follows from (24), the third from the substitution $\lambda_S = \Lambda_S(t)$ and (7), which states that $\frac{d\lambda_S}{dt} = -\rho \mu(\lambda_S)$, and the final equality from Lemma 2. $\square$

Remark 4. Recall that $\mathcal{E}(\Lambda_R(0))$ is interpreted as the average school enrollment. This must equal the ratio of students to schools ρ times the fraction of students who are assigned $F_R(\infty)$. Meanwhile, the quantity $(1 - \mathcal{L}(k))(1 - F_R(k|\infty))$ gives the fraction of students who express interest in at least $k + 1$ schools (they list more than k schools and do not get into any of their first k choices). Therefore, by Lemma 1, the right side of (22) is the average number of schools at which a student expresses interest, times the ratio of students to schools, which must equal schools' average number of interested students $\Lambda_R(0)$.

Proof of Proposition 1. The function Λ_S is given as the solution to a one-dimensional differential equation with initial condition $\Lambda_S(1) = 0$. By the Picard Lindelöf theorem, to show that this has a unique solution, it suffices to show that $\mu(\mathcal{V}(\lambda))$ is Lipschitz in λ . This holds because $|\frac{d}{d\lambda} \mu(\mathcal{V}(\lambda))| = |\mathcal{V}'(\lambda)| |\mu'(\mathcal{V}(\lambda))|$, and Lemmas 2 and 4 imply that $\mathcal{V}'$ and μ' are bounded.

Turning to Λ_I , by (9), it suffices to show that there is a unique solution $\lambda = \Lambda_I(0)$ to (10). Because $\lambda = 0$ is not a solution, we can multiply both sides by $\mathcal{E}(\lambda)/\lambda$ and look for solutions to

$$\mathcal{E}(\lambda) = \rho \mathcal{A}(\lambda) \mu(\mathcal{A}(\lambda)).$$

The left side is zero at $\lambda = 0$ and strictly increasing. The right side is equal to ρ at $\lambda = 0$ and by Lemmas 3 and 4 is strictly decreasing. It approaches zero as $\lambda \rightarrow \infty$, so there is a unique intersection point. $\square$

A.1. Proofs of Theorem 1 and Proposition 2.

Remark 5. *This remark provides interpretation for the quantities used in the proof of Theorem 1. The quantity λ'_R can be interpreted as the expected amount of additional interest triggered by the addition of a new student to the market under priority rule R . Note that this may be higher than the number of schools at which this student expresses interest, because the student may cause others to be rejected. Because schools on each student's list are sampled iid, there is a one-to-one correspondence between the total interest λ_R and enrollment $\mathcal{E}(\lambda_R)$. Thus, the strategy is to show that whenever the two methods have matched the same number of students so far, then the number of new proposals (or equivalently, the number of new matches) triggered by the addition of one student is higher under one procedure than the other.*

Proof of Theorem 1. Because $\mathcal{E}(\cdot)$ is monotonic, it suffices to compare $\lambda_I = \Lambda_I(0)$ to $\lambda_S = \Lambda_S(0)$. We do this by considering these values as functions of ρ (for fixed $\mathcal{L}$, $\mathcal{T}$, and $\mathcal{D}$).

Because $\lambda_I(0) = \lambda_S(0) = 0$ and both λ_I and λ_S are continuous and differentiable functions of ρ , to show that $\lambda_I(\rho) \geq \lambda_S(\rho)$ for all ρ , it suffices to show that

$$\lambda_I(\rho_I) = \lambda_S(\rho_C) \Rightarrow \lambda'_I(\rho_I) > \lambda'_S(\rho_C).$$

Similarly, to show that $\lambda_I(\rho) \leq \lambda_S(\rho)$ for all ρ , it suffices to show that

$$\lambda_I(\rho_I) = \lambda_S(\rho_C) \Rightarrow \lambda'_I(\rho_I) < \lambda'_S(\rho_C).$$

We now compute λ'_I and λ'_S . Differentiating (10) with respect to ρ (and dropping the implicit dependence of λ_I and λ'_I on ρ), we get

$$(25) \quad \lambda'_I = \mu(\mathcal{A}(\lambda_I)) + \rho \mu'(\mathcal{A}(\lambda_I)) \mathcal{A}'(\lambda_I) \lambda'_I.$$

From Lemma 3 and (10), it follows that that

$$\rho \cdot \mathcal{A}'(\lambda_I) = \rho \frac{\mathcal{V}(\lambda_I) - \mathcal{A}(\lambda_I)}{\lambda_I} = \frac{\mathcal{V}(\lambda_I) - \mathcal{A}(\lambda_I)}{\mu(\mathcal{A}(\lambda_I))}.$$

Substituting this expression into (25) and solving for λ'_I , we see that the derivative of λ_I with respect to ρ is

$$(26) \quad \lambda'_I = \frac{\mu(\mathcal{A}(\lambda_I))}{1 - (\mathcal{V}(\lambda_I) - \mathcal{A}(\lambda_I)) \frac{\mu'(\mathcal{A}(\lambda_I))}{\mu(\mathcal{A}(\lambda_I))}}.$$

Meanwhile, we claim that

$$(27) \quad \lambda'_S = -\frac{1}{\rho} \Lambda'_S(0) = \mu(\mathcal{V}(\lambda_S)).$$

This can be seen by noting that if we view Λ_S as a function of t and ρ , then for $\bar{\rho} > \rho$, $\lambda_S(\rho) = \Lambda_S(0, \rho) = \Lambda_S(1 - \rho/\bar{\rho}, \bar{\rho})$ (both quantities represent the total interest expressed by the first ρ students). Differentiating this equation with respect to ρ yields (27).

It follows from (26) and (27) that if $\lambda_I(\rho_I) = \lambda_S(\rho_C) = \lambda$, then (letting $\mathcal{V} = \mathcal{V}(\lambda)$, $\mathcal{A} = \mathcal{A}(\lambda)$)

$$\begin{aligned} \lambda'_S(\rho_C) < \lambda'_I(\rho_I) &\Leftrightarrow \mu(\mathcal{V}) < \frac{\mu(\mathcal{A})}{1 - (\mathcal{V} - \mathcal{A}) \frac{\mu'(\mathcal{A})}{\mu(\mathcal{A})}} \\ &\Leftrightarrow \mu(\mathcal{V}) - \mu(\mathcal{A}) < (\mathcal{V} - \mathcal{A}) \mu'(\mathcal{A}) \frac{\mu(\mathcal{V})}{\mu(\mathcal{A})} \\ &\Leftrightarrow \frac{\mu(\mathcal{V}) - \mu(\mathcal{A})}{\mu(\mathcal{V}) \mu(\mathcal{A})} < (\mathcal{A} - \mathcal{V}) f'(\mathcal{A}). \\ &\Leftrightarrow \frac{f(\mathcal{A}) - f(\mathcal{V})}{\mathcal{A} - \mathcal{V}} < f'(\mathcal{A}), \end{aligned}$$

where we have used the fact that $f'(\alpha) = -\mu'(\alpha)/\mu(\alpha)^2$. Because $\mathcal{V} < \mathcal{A}$ by Lemma 3, this holds if f is convex; if f is concave, the inequality reverses. □

Proof of Proposition 2. Differentiating the function f we obtain

$$f''(\alpha) = \frac{2\mu'(\alpha)^2 - \mu(\alpha)\mu''(\alpha)}{\mu(\alpha)^3}.$$

Thus, to determine convexity or concavity of f , it suffices to study the sign of $2\mu'(\alpha)^2 - \mu(\alpha)\mu''(\alpha)$. Straightforward algebra reveals that

$$(28) \quad f''(\alpha)\mu(\alpha)^3 = 2\mu'(\alpha)^2 - \mu(\alpha)\mu''(\alpha) = \sum_{k=0}^{\infty} q_k(1-\alpha)^k,$$

where

$$q_{k-2} = \frac{1}{2} \sum_{i=0}^k (4i(k-i) - i(i-1) - (k-i)(k-i-1)) (1 - \mathcal{L}(i))(1 - \mathcal{L}(k-i)).$$

For $k \in \mathbb{N}$, define $\bar{\mathcal{L}}(k) = 1 - \mathcal{L}(k)$ to be the fraction of students listing more than k schools. For $i \leq k$, define

$$\begin{aligned} r_i^k &= 4i(k-i) - i(i-1) - (k-i)(k-i-1), \\ s_i^k &= \bar{\mathcal{L}}(i)\bar{\mathcal{L}}(k-i), \end{aligned}$$

so that

$$q_{k-2} = \frac{1}{2} \sum_{i=0}^k r_i^k s_i^k.$$

Noting that $r_i^k = r_{k-i}^k$ and $s_i^k = s_{k-i}^k$, we can express q_{k-2} as follows:

$$(29) \quad q_{k-2} = \sum_{i=0}^{\lfloor k/2 \rfloor} \tilde{r}_i^k s_i^k,$$

where we define $\tilde{r}_i^k = r_i^k$ if $i < k/2$ and $\tilde{r}_i^k = r_i^k/2$ if $i = k/2$.

We will show that

- i) The $\tilde{r}_i^k$ “cross zero” once and sum to zero.

ii) If $\mathcal{L}$ has a weakly increasing hazard rate, then s_i^k is increasing on $\{0, 1, \dots, \lfloor k/2 \rfloor\}$.

If $\mathcal{L}$ has a weakly decreasing hazard rate, then s_i^k is decreasing on $\{0, 1, \dots, \lfloor k/2 \rfloor\}$.

From i), ii) and (29), Lemma 6 implies that if $\mathcal{L}$ has a weakly increasing hazard rate, then each q_k is non-negative; if $\mathcal{L}$ has a weakly decreasing hazard rate, then each q_k is non-positive. By (28) the proposition immediately follows.

All that remains is to establish i) and ii). Note that

$$(30) \quad r_i^k = 6i(k-i) - k(k-1),$$

which is a quadratic in i . It is negative at $i = 0$ and obtains its maximal value at $k/2$. It follows that the sequence $\{r_i^k\}_{i=1}^{\lfloor k/2 \rfloor}$ is initially negative and later non-negative, so the same is true of $\tilde{r}_i^k$. Furthermore, the definition of $\tilde{r}_i^k$ implies that

$$\sum_{i=0}^{\lfloor k/2 \rfloor} \tilde{r}_i^k = \sum_{i=0}^k r_i^k = 0,$$

where the final equality follows from (30) by applying standard formulas for $\sum_{i=0}^k i$ and $\sum_{i=0}^k i^2$.

Point ii) follows because for $i \geq 1$,

$$(31) \quad s_{i-1}^k \leq s_i^k \Leftrightarrow \frac{\bar{\mathcal{L}}(k-i+1)}{\bar{\mathcal{L}}(k-i)} \leq \frac{\bar{\mathcal{L}}(i)}{\bar{\mathcal{L}}(i-1)} \Leftrightarrow 1 - \frac{\bar{\mathcal{L}}(i)}{\bar{\mathcal{L}}(i-1)} \leq 1 - \frac{\bar{\mathcal{L}}(k-i+1)}{\bar{\mathcal{L}}(k-i)}.$$

If $i \leq \lfloor k/2 \rfloor$, then $i < k-i+1$, so if $\mathcal{L}$ has an increasing hazard rate, then all inequalities in (31) hold, and if $\mathcal{L}$ has a decreasing hazard rate, then all inequalities reverse. $\square$

Lemma 6. *Let $\{r_i\}_{i=0}^k$ be a sequence of real numbers with mean zero that “crosses zero” once. That is, $\sum_{i=0}^k r_i = 0$ and for some $j < k$, it holds that $r_i < 0$ for $i \leq j$ and $r_i \geq 0$ for $j < i \leq k$. Let $\{s_i\}_{i=0}^k$ be a sequence of non-negative numbers. If s_i is weakly increasing, then $\sum_{i=0}^k r_i s_i \geq 0$, and if s_i is weakly decreasing, then $\sum_{i=0}^k r_i s_i \leq 0$.*

Proof. If s_i is weakly increasing, then

$$\sum_{i=0}^k r_i s_i = \sum_{i=0}^j r_i s_i + \sum_{i=j+1}^k r_i s_i \geq \sum_{i=0}^j r_i s_j + \sum_{i=j+1}^k r_i s_j = s_j \sum_{i=1}^k r_i = 0,$$

where the inequality follows because the first sum (consisting of negative terms) has been made “less negative” and the second sum (consisting of non-negative terms) has been made “more positive.” Analogous logic holds when s_i is weakly decreasing. $\square$

A.2. Proof of Theorem 2. The proof of Theorem 2 follows from Lemmas 7 and 8. Lemma 7 says that $F_I(\cdot|\infty)$ and $F_S(\cdot|\infty)$ have at most one crossing point, implying that either there is one crossing point or one dominates the other. Lemma 8 states that it is not possible for $F_I(\cdot|\infty)$ to dominate $F_S(\cdot|\infty)$, or vice versa.

Lemma 7. *If $F_I(k|\infty) \geq F_S(k|\infty)$, then $F_I(k'|\infty) > F_S(k'|\infty)$ for all $k' > k$.*

Proof. Recall that

$$F_I(k|\infty) = 1 - (1 - \alpha(\Lambda_I(0)))^k, \quad F_S(k|\infty) = 1 - \int_0^1 (1 - \mathcal{V}(\Lambda_S(t)))^k dt.$$

It follows that for $k' \geq k$,

$$\begin{aligned} 1 - F_I(k'|\infty) &= (1 - F_I(k|\infty))^{k'/k} \\ &< (1 - F_S(k|\infty))^{k'/k} \\ &= \left(\int_0^1 (1 - \mathcal{V}(\Lambda_S(t)))^k dt \right)^{k'/k} \\ &\leq \int_0^1 (1 - \mathcal{V}(\Lambda_S(t)))^{k'} dt \\ &= 1 - F_S(k'|\infty), \end{aligned}$$

where the first inequality follows because $F_I(k|\infty) > F_S(k|\infty)$, and the second by Jensen's inequality (see Fact 2). □

Fact 2 (Jensen's Inequality). *Let $f, g : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, fix $A \subseteq \mathbb{R}_+$, and suppose that $\int_A f(t) dt < \infty$. Then*

- *If g is convex, $\int_A g(f(t)) dt \geq g(\int_A f(t) dt)$.*
- *If g is concave, $\int_A g(f(t)) dt \leq g(\int_A f(t) dt)$.*

Lemma 8. *There exists $k \in \mathbb{N}$ such that $F_I(k|\infty) > F_S(k|\infty)$ if and only if there exists $k' \in \mathbb{N}$ such that $F_I(k|\infty) < F_S(k|\infty)$.*

Proof. Let $L \subseteq \mathbb{N}$ be the support of the list length distribution. Seeking a contradiction, suppose that one of the procedures dominates the other. That is, for $R, \tilde{R} \in \{I, S\}$,

$$(32) \quad F_R(k|\infty) \geq F_{\tilde{R}}(k|\infty) \text{ for all } k \in \mathbb{N}, \text{ with strict inequality for some } k \in L.$$

Because $F_R(k|\infty) = F_R(\infty|k)$, (16) and (32) jointly imply that $F_R(\infty) > F_{\tilde{R}}(\infty)$.

But (22) and (32) imply that $\Lambda_R(0) < \Lambda_{\tilde{R}}(0)$, and therefore that $\mathcal{E}(\Lambda_R(0)) < \mathcal{E}(\Lambda_{\tilde{R}}(0))$. By (23), this implies $F_R(\infty) < F_{\tilde{R}}(\infty)$, which is a contradiction. Thus, (32) cannot hold. □

A.3. Proofs of Theorem 3 and Proposition 3. Our first step is to show that if the mass of students exceeds the total mass of seats, then under either lottery procedure the total amount of interest grows linearly in the length of student lists.

Lemma 9. *Suppose that $\mathcal{L}(k) = \mathbf{1}(k \geq \ell)$ (all students list ℓ schools) and $\rho > \sum_{\tau \in \mathcal{T}} \mathcal{D}_\tau C_\tau$. Then*

$$(33) \quad \min(\Lambda_I(0), \Lambda_S(0)) \geq \ell(\rho - \sum_{\tau \in \mathcal{T}} \mathcal{D}_\tau C_\tau).$$

Proof. We start by noting that when all lists have length ℓ ,

$$(34) \quad \mu(\alpha) = \sum_{k=0}^{\infty} (1 - \mathcal{L}(k))(1 - \alpha)^k = \sum_{k=0}^{\ell-1} (1 - \alpha)^k \geq \ell(1 - \alpha)^\ell.$$

It follows from (10) that

$$(35) \quad \Lambda_I(0) = \rho\mu(\mathcal{A}(\Lambda_I(0))) \geq \ell\rho(1 - \mathcal{A}(\Lambda_I(0)))^\ell = \ell\rho(1 - F_I(\infty)) = \ell(\rho - \mathcal{E}(\Lambda_I(0))).$$

The inequality follows from (34), the subsequent equality from the definition of F_I in (11) and (16), and the final equality from Lemma 5. Similarly (6) and (7) imply that

$$(36) \quad \begin{aligned} \Lambda_S(0) &= \int_0^1 \rho\mu(\mathcal{V}(\Lambda_S(t)))dt. \\ &\geq \ell\rho \int_0^1 (1 - \mathcal{V}(\Lambda_S(t)))^\ell dt \\ &= \ell\rho(1 - F_S(\infty)) = \ell(\rho - \mathcal{E}(\Lambda_S(0))). \end{aligned}$$

The inequality uses (34), and the final line uses the definition of F_S in (8) and (16) and Lemma 5. The result follows Lemma 2, which states that $\mathcal{E}(\cdot)$ is upper-bounded by $\sum_{\tau \in \mathcal{T}} \mathcal{D}_\tau C_\tau$. $\square$

Remark 6. *The bound in Lemma 9 has a natural interpretation: capacity constraints imply that the mass of unassigned students must be at least $\rho - \sum_{\tau \in \mathcal{T}} \mathcal{D}_\tau C_\tau$, and each unassigned student expresses interest in all ℓ schools on her list.*

Proof of Theorem 3. We begin with the statement about independent lotteries. The fraction of students who receive their top choice is $F_I(1) = \mathcal{A}(\Lambda_I(0))$, and by Lemma 5 the fraction who are matched is $F_I(\infty) = \frac{1}{\rho}\mathcal{E}(\Lambda_I(0))$. By (3), it follows that

$$\frac{F_I(1)}{F_I(\infty)} = \frac{\rho \cdot \mathcal{A}(\Lambda_I(0))}{\mathcal{E}(\Lambda_I(0))} = \frac{\rho}{\Lambda_I(0)}.$$

Lemma 9 implies that for $\rho > \sum_{\tau} \mathcal{D}_\tau C_\tau$, this ratio can be made arbitrarily small by letting $\mathcal{L}(k) = \mathbf{1}(k \geq \ell)$ for sufficiently large ℓ .

We now turn to the statement for the single lottery, using the shorthand $\lambda_S = \Lambda_S(0)$. By Lemma 5, the fraction of students who are assigned is

$$F_S(\infty) = \frac{1}{\rho}\mathcal{E}(\lambda_S) = \frac{1}{\rho} \int_0^{\lambda_S} \mathcal{V}(\lambda) d\lambda,$$

where the second equality uses Lemma 2. The fraction of students who receive their top choice is

$$F_S(1) = \int_0^1 \mathcal{V}(\Lambda_S(t)) dt = \frac{1}{\rho} \int_0^{\lambda_S} \frac{\mathcal{V}(\lambda)}{\mu(\mathcal{V}(\lambda))} d\lambda \geq \frac{1}{\rho} \int_0^{\lambda_S} \mathcal{V}(\lambda)^2 d\lambda,$$

where we use the fact that $1/\mu(\mathcal{V}(\lambda)) \geq \mathcal{V}(\lambda)$ by Lemma 4. It follows that

$$(37) \quad \frac{F_S(1)}{F_S(\infty)} \geq \frac{\int_0^{\lambda_S} \mathcal{V}(\lambda)^2 d\lambda}{\int_0^{\lambda_S} \mathcal{V}(\lambda) d\lambda} \geq \frac{\int_0^{\infty} \mathcal{V}(\lambda)^2 d\lambda}{\int_0^{\infty} \mathcal{V}(\lambda) d\lambda}.$$

The final inequality follows because

$$\frac{d}{d\lambda_S} \log \left(\frac{\int_0^{\lambda_S} \mathcal{V}(\lambda)^2 d\lambda}{\int_0^{\lambda_S} \mathcal{V}(\lambda) d\lambda} \right) = \frac{\mathcal{V}(\lambda_S)^2}{\int_0^{\lambda_S} \mathcal{V}(\lambda)^2 d\lambda} - \frac{\mathcal{V}(\lambda_S)}{\int_0^{\lambda_S} \mathcal{V}(\lambda) d\lambda} < 0,$$

with the inequality following from the fact that $\mathcal{V}(\cdot)$ is decreasing.

We now show that the inequalities in (37) can be made tight. For $\ell \in \mathbb{N}$, take $\mathcal{L}(k) = \mathbf{1}(k \geq \ell)$, meaning that all students list exactly ℓ schools. Then

$$\begin{aligned}
 F_S(1) - \frac{1}{\rho} \int_0^{\lambda_S} \mathcal{V}(\lambda)^2 d\lambda &= \frac{1}{\rho} \int_0^{\lambda_S} \mathcal{V}(\lambda) (1/\mu(\mathcal{V}(\lambda)) - \mathcal{V}(\lambda)) d\lambda \\
 &\leq \frac{1}{\rho\ell} \int_0^{\lambda_S} \mathcal{V}(\lambda) d\lambda \\
 (38) \qquad \qquad \qquad &\leq \frac{1}{\rho\ell} \sum_{\tau \in \mathcal{T}} \mathcal{D}_\tau C_\tau.
 \end{aligned}$$

The second inequality follows by Lemma 2, and the first because when $\mathcal{L}(k) = \mathbf{1}(k \geq \ell)$, we have that for $\alpha \in (0, 1]$,

$$1/\mu(\alpha) - \alpha = \frac{\alpha(1 - \alpha)^\ell}{1 - (1 - \alpha)^\ell},$$

which is positive, decreasing in α , and approaches $1/\ell$ as $\alpha \rightarrow 0$. Note that (38) implies that the first inequality in (37) becomes tight as $\ell \rightarrow \infty$. Furthermore, Lemma 9 states that if $\rho > \sum_{\tau \in \mathcal{T}} \mathcal{D}_\tau C_\tau$, then $\lambda_S \rightarrow \infty$ as $\ell \rightarrow \infty$, implying that the second inequality in (37) becomes tight. $\square$

In light of Theorem 3, given a school type space $\mathcal{T}$ and a distribution $\mathcal{D}$ on $\mathcal{T}$, define

$$(39) \qquad \qquad \qquad LB(\mathcal{T}, \mathcal{D}) = \frac{\int_0^\infty \mathcal{V}(\lambda)^2 d\lambda}{\int_0^\infty \mathcal{V}(\lambda) d\lambda}.$$

Note that the dependence on $\mathcal{T}$ and $\mathcal{D}$ is implicit, through the function $\mathcal{V}(\cdot)$.

We now turn to the proof of Proposition 3. Our first step is Lemma 10, which establishes that when schools have identical capacities, the distribution $\mathcal{D}$ that minimizes the bound in (39) places all of its mass on the extreme points of $\mathcal{T}$. That is, the worst-case distribution of popularity is to have two types of schools: one popular, and the other unpopular.

Lemma 10. *Fix $\mathcal{T}$ such that $C_\tau = C \in \mathbb{N}$ for all $\tau \in \mathcal{T}$, and define*

$$(40) \qquad \qquad \qquad \underline{p} = \min_{\tau \in \mathcal{T}} p_\tau, \qquad \bar{p} = \max_{\tau \in \mathcal{T}} p_\tau.$$

Let $\mathcal{D}^(\mathcal{T})$ be the distribution that places probability β on $(\underline{p}, C)$ and probability $1 - \beta$ on $(\bar{p}, C)$, where $\beta \underline{p} + (1 - \beta) \bar{p} = 1$. Then $LB(\mathcal{T}, \mathcal{D}^*(\mathcal{T})) = \inf_{\mathcal{D}} LB(\mathcal{T}, \mathcal{D})$.*

Proof. In this proof, it will be helpful to make the dependence of $\mathcal{V}$ and $\mathcal{E}$ on $\mathcal{D}$ explicit. Thus, we write $\mathcal{V}(\lambda, \mathcal{D})$ and $\mathcal{E}(\lambda, \mathcal{D})$ in place of $\mathcal{V}(\lambda)$ and $\mathcal{E}(\lambda)$. Because $\mathcal{T}$ is fixed, we write $\mathcal{D}^*$ for $\mathcal{D}^*(\mathcal{T})$.

By Lemma 2, for any $\mathcal{D}$ we have

$$\int_0^\infty \mathcal{V}(\lambda, \mathcal{D}) d\lambda = \lim_{\lambda \rightarrow \infty} \mathcal{E}(\lambda, \mathcal{D}) = \sum_{\tau \in \mathcal{T}} \mathcal{D}_\tau C_\tau.$$

When $C_\tau = C$ for all τ , it follows that this expression is C , so minimizing the lower bound in (39) is equivalent to minimizing the numerator $\int_0^\infty \mathcal{V}(\lambda, \mathcal{D})^2 d\lambda$.

We show that if $\mathcal{D}$ places positive mass $q > 0$ on a school type $\tau' \in \mathcal{T}$ with $p_{\tau'} = p \in (\underline{p}, \bar{p})$, then moving this mass to the extreme points $\underline{p}$ and $\bar{p}$ decreases LB . Formally, let $\tilde{\mathcal{D}}$ be the modification of $\mathcal{D}$ in which the probability of τ' under $\tilde{\mathcal{D}}$ is reduced to zero, and the probability of $(\underline{p}, C)$ and $(\bar{p}, C)$ are increased by $q\beta$ and $q(1 - \beta)$, respectively. The definition of β ensures that $\tilde{\mathcal{D}}$ remains a

probability distribution and that average popularity $\sum_{\tau} \tilde{\mathcal{D}}_{\tau} p_{\tau} = 1$. We will show that

$$(41) \quad \int_0^{\infty} \mathcal{V}(\lambda, \tilde{\mathcal{D}})^2 d\lambda \leq \int_0^{\infty} \mathcal{V}(\lambda, \mathcal{D})^2 d\lambda.$$

Note that for any $\mathcal{D}$, $\mathcal{V}(\lambda, \mathcal{D})$ is one at $\lambda = 0$ and decreases to zero as $\lambda \rightarrow \infty$. Our first step is to show that the random variable X that has CDF $1 - \mathcal{V}(\cdot, \mathcal{D})$ second order stochastically dominates that the random variable $\tilde{X}$ with CDF $1 - \mathcal{V}(\cdot, \tilde{\mathcal{D}})$. That is, for any $\lambda \in \mathbb{R}_+$,

$$(42) \quad \int_0^{\lambda} \mathcal{V}(x, \mathcal{D}) - \mathcal{V}(x, \tilde{\mathcal{D}}) dx = \mathcal{E}(\lambda, \mathcal{D}) - \mathcal{E}(\lambda, \tilde{\mathcal{D}}) \geq 0,$$

where the equality above uses Lemma 2. By (13) we have $\mathcal{E}(\lambda, \mathcal{D}) = \mathbb{E}_{\tau \sim \mathcal{D}}[\varepsilon(p_{\tau} \lambda, C)]$. Lemma 2 implies that ε is concave in its first argument, from which it follows that if we let $\tau \sim \mathcal{D}$ and $\tilde{\tau} \sim \tilde{\mathcal{D}}$, then $\mathbb{E}[p_{\tau}] = \mathbb{E}[p_{\tilde{\tau}}]$ and by Jensen's inequality (42) holds.

Having established second order stochastic dominance, (41) follows from a chain of inequalities:

$$\begin{aligned} \int_0^{\infty} \mathcal{V}(\lambda, \tilde{\mathcal{D}})^2 d\lambda &= - \int_0^{\infty} \mathcal{E}(\lambda, \tilde{\mathcal{D}}) \mathcal{V}'(\lambda, \tilde{\mathcal{D}}) d\lambda \\ &\leq - \int_0^{\infty} \mathcal{E}(\lambda, \tilde{\mathcal{D}}) \mathcal{V}'(\lambda, \mathcal{D}) d\lambda \\ &= \int_0^{\infty} \mathcal{V}(\lambda, \mathcal{D}) \mathcal{V}(\lambda, \tilde{\mathcal{D}}) d\lambda \\ &= - \int_0^{\infty} \mathcal{E}(\lambda, \mathcal{D}) \mathcal{V}'(\lambda, \tilde{\mathcal{D}}) d\lambda \\ &\leq - \int_0^{\infty} \mathcal{E}(\lambda, \mathcal{D}) \mathcal{V}'(\lambda, \mathcal{D}) d\lambda \\ &= \int_0^{\infty} \mathcal{V}(\lambda, \mathcal{D})^2 d\lambda. \end{aligned}$$

The notation $\mathcal{V}'(\lambda, \mathcal{D})$ denotes the derivative of $\mathcal{V}$ with respect to λ . All equalities above follow from integration by parts, using the fact that $\mathcal{E}'(\lambda) = \mathcal{V}(\lambda)$ (Lemma 2), $\mathcal{E}(0) = 0$, and $\mathcal{V}(\infty) = 0$. The inequalities follow from second-order stochastic dominance, which implies that for any concave increasing function u ,

$$- \int_0^{\infty} u(\lambda) \mathcal{V}'(\lambda, \tilde{\mathcal{D}}) d\lambda = \mathbb{E}[u(\tilde{X})] \leq \mathbb{E}[u(X)] = - \int_0^{\infty} u(\lambda) \mathcal{V}'(\lambda, \mathcal{D}) d\lambda.$$

In the first inequality, we use $u(\cdot) = \mathcal{E}(\cdot, \tilde{\mathcal{D}})$, and in the second we use $u(\cdot) = \mathcal{E}(\cdot, \mathcal{D})$. Both functions are concave and increasing by Lemma 2. $\square$

Having determined the worst-case distribution $\mathcal{D}$ for a given $\mathcal{T}$, we now determine the worst-case type space $\mathcal{T}$ satisfying the constraints that all schools have equal capacity and no school is more than r times as popular as any other. Given $r \geq 1$ and $C \in \mathbb{N}$, define

$$(43) \quad T(r, C) = \{\mathcal{T} : C_{\tau} = C \text{ for all } \tau \in \mathcal{T}, \text{ and } \underline{p} \leq 1 \leq \bar{p} \leq r\underline{p}\}.$$

$$(44) \quad LBT(r, C) = \inf_{\mathcal{T} \in T(r, C)} LB(\mathcal{T}, \mathcal{D}^*(\mathcal{T})).$$

The following Lemma shows that the worst case can be achieved by a $\mathcal{T}$ for which some schools have popularity $\sqrt{r}$ and others have popularity $1/\sqrt{r}$. Proposition 3 corresponds to the statement about $LBT(r, 1)$.

Lemma 11. *Fix $r \geq 1$ and $C \in \mathbb{N}$, and define $\mathcal{T}^* = \{(\sqrt{r}, C), (1/\sqrt{r}, C)\} \in T(r, C)$. Then*

$$(45) \quad LBT(r, C) = LB(\mathcal{T}^*, \mathcal{D}^*(\mathcal{T}^*)),$$

from which it follows that

$$(46) \quad LBT(r, 1) = \frac{\sqrt{r}}{1+r}, \quad \lim_{C \rightarrow \infty} LBT(r, C) = 2 \frac{\sqrt{r} - 1}{r - 1}.$$

Proof. For any $\mathcal{T} \in T(r, C)$ it follows from Lemma 2 and (12) that

$$LB(\mathcal{T}, \mathcal{D}^*(\mathcal{T})) = \frac{1}{C} \int_0^\infty \nu(\lambda)^2 d\lambda = \frac{1}{C} \int (\beta \bar{p} \nu(\bar{p}\lambda, C) + (1 - \beta) \underline{p} \nu(\underline{p}\lambda, C))^2 d\lambda.$$

Expanding this expression yields

$$\frac{1}{C} \left(\beta^2 \bar{p}^2 \int \nu(\bar{p}\lambda, C)^2 d\lambda + 2\beta(1 - \beta) \bar{p} \underline{p} \int \nu(\bar{p}\lambda, C) \nu(\underline{p}\lambda, C) d\lambda + (1 - \beta)^2 \underline{p}^2 \int \nu(\underline{p}\lambda, C)^2 d\lambda \right).$$

Applying the u -substitution $\lambda = Cx/\bar{p}$ to the first two integrals and $\lambda = Cx/\underline{p}$ to the third, we get

$$(47) \quad LB(\mathcal{T}, \mathcal{D}^*(\mathcal{T})) = (\beta^2 \bar{p} + (1 - \beta)^2 \underline{p}) \int_0^\infty \nu(Cx, C)^2 dx + 2\beta(1 - \beta) \bar{p} \underline{p} \int_0^\infty \nu(Cx, C) \nu(Cx \underline{p}/\bar{p}, C) dx.$$

To find $LBT(r, C)$ we must choose $\underline{p}$ and $\bar{p}$ to minimize this expression. The argument used in Lemma 10 implies that $LB(\mathcal{T}, \mathcal{D}^*(\mathcal{T}))$ is decreasing in $\bar{p}$ for fixed $\underline{p}$, so the worst case $\mathcal{T} \in T(r, C)$ satisfies $\bar{p} = r\underline{p}$. Combining this with the equation $\beta \bar{p} + (1 - \beta) \underline{p} = 1$ that defines β in Lemma 10 and solving for $\bar{p}$, we get

$$(48) \quad \bar{p} = \frac{r}{\beta r + 1 - \beta}$$

We find $LBT(r, C)$ by minimizing over $\bar{p}$, or equivalently over β . We adopt the latter formulation. Define

$$(49) \quad \gamma_1(\beta, r) = \beta^2 \bar{p} + (1 - \beta)^2 \bar{p}/r = \frac{\beta^2 r + (1 - \beta)^2}{\beta r + 1 - \beta},$$

$$(50) \quad \gamma_2(\beta, r) = 2\beta(1 - \beta) \bar{p}/r = \frac{2\beta(1 - \beta)}{\beta r + 1 - \beta},$$

$$(51) \quad h(\beta, r, C) = \gamma_1(\beta, r) \int_0^\infty \nu(Cx, C)^2 dx + \gamma_2(\beta, r) \int_0^\infty \nu(Cx, C) \nu(Cx/r, C) dx.$$

Then by (47),

$$(52) \quad LBT(r, C) = \min_{\beta \in [0, 1]} h(\beta, r, C)$$

We fix r and C , and drop dependence of γ_1, γ_2, h on these parameters. Because h is differentiable, its minimum is either at $\beta \in \{0, 1\}$ or a solution to $h'(\beta) = 0$. The value at the end points is

$$h(0) = h(1) = \int_0^\infty \nu(Cx, C)^2 dx.$$

We now seek solutions to $h'(\beta) = 0$. To simplify the calculations, we note that (49) and (50) imply

$$(53) \quad \gamma_2(\beta) = \frac{2}{1+r}(1 - \gamma_1(\beta)).$$

From this and (51), it follows that

$$(54) \quad h'(\beta) = \gamma_1'(\beta) \left(\int_0^\infty \nu(Cx, C)^2 dx - \frac{2}{1+r} \int_0^\infty \nu(Cx, C) \nu(Cx/r, C) dx \right)$$

If the term in parentheses is nonzero, this implies that $h'(\beta) = 0$ if and only if $\gamma_1'(\beta) = 0$, which by (53) occurs if and only if $\gamma_2'(\beta) = 0$ (if the term in parentheses is zero, then h is constant). But

$$\gamma_2'(\beta) = -2 \frac{(r-1)\beta^2 + 2\beta - 1}{((r-1)\beta + 1)^2},$$

from which it follows that the only solution to $\gamma_2'(\beta) = 0$ on $[0, 1]$ is $\beta = \frac{\sqrt{r}-1}{r-1}$, which corresponds to $\bar{p} = \sqrt{r}$. One can verify that indeed this is a minimum, not a maximum (that is, $h(\frac{\sqrt{r}-1}{r-1})$ is less than $h(0) = h(1)$). This establishes (45). From this and (47), it follows that

$$LBT(r, C) = 2 \left(\frac{\sqrt{r}-1}{r-1} \right)^2 \left(\sqrt{r} \int \nu(Cx, C)^2 dx + \int \nu(Cx, C) \nu(Cx/r, C) dx \right).$$

For $C = 1$, $\nu(Cx, C) = e^{-x}$, and the expression above evaluates to $\sqrt{r}/(1+r)$.

As C grows, $\nu(Cx, C)$ converges pointwise to $g(x) = \mathbf{1}(x < 1)$. Because $\nu(Cx, C)$ is pointwise less than e^{1-x} , by the dominated convergence theorem we have

$$\lim_{C \rightarrow \infty} LBT(r, C) = 2 \left(\frac{\sqrt{r}-1}{r-1} \right)^2 \left(\sqrt{r} \int g(x)^2 dx + \int g(x)g(x/r) dx \right) = 2 \frac{\sqrt{r}-1}{r-1},$$

where the final equality follows because $g(x)^2 = g(x) = g(x)g(x/r)$ and therefore both integrals above are equal to one. $\square$

A.4. Comparison of Unmatched Agents. The following result considers a special case: a one-to-one market with equally popular schools in which student list lengths follow a Poisson distribution with mean ℓ .¹⁴ If there are fewer students than seats ($\rho < 1$), then the number of unassigned students decays exponentially in ℓ under both procedures, but at different rates. The difference in rates is especially large if ρ is close to one. For example, if $\rho = 0.9$, then $1 - F_I(\infty) \approx e^{-0.4\ell}$, whereas $1 - F_S(\infty) \approx e^{-0.1\ell}$. If $\rho = 0.96$ then $1 - F_I(\infty) \approx e^{-0.3\ell}$, whereas $1 - F_S(\infty) \approx e^{-0.04\ell}$. When the market is balanced ($\rho = 1$), the number of unassigned students under independent lotteries decays exponentially in $\sqrt{\ell}$, whereas under a single lottery it is approximately $\log(2)/\ell$.

Proposition 4. *Suppose that schools are equally popular and have a single seat, and the list length distribution is Poisson with mean ℓ . If $\rho = 1$, then*

$$1 - F_I(\infty) = e^{-\sqrt{\ell F_I(\infty)}} \sim e^{-\sqrt{\ell}}, \quad 1 - F_S(\infty) = \log(2 - e^{-\ell})/\ell \sim \log(2)/\ell.$$

¹⁴It is possible to derive similar results when all students submit lists of length ℓ , though in that case it is not possible to derive a closed-form expression for $F_S(\infty)$ in terms of market primitives.

If $\rho < 1$, then as $\ell \rightarrow \infty$,

$$(55) \quad \frac{1}{\ell} \log(1 - F_I(\infty)) = \frac{\rho F_I(\infty)}{\log(1 - \rho F_I(\infty))} \rightarrow \frac{\rho}{\log(1 - \rho)}.$$

$$(56) \quad \frac{1}{\ell} \log(1 - F_S(\infty)) = \frac{1}{\ell} \log \left(1 - \frac{1}{\rho} + \frac{1}{\rho \ell} \log(1 + e^{\ell(1-\rho)} - e^{-\ell\rho}) \right) \rightarrow -(1 - \rho).$$

Proof. With a single lottery, we use Lemma (2) and (7) to note that

$$(57) \quad \frac{d}{dt} \mathcal{E}(\Lambda_S(t)) = \mathcal{V}(\Lambda_S(t)) \Lambda'_S(t) = -\rho \mathcal{V}(\Lambda_S(t)) \mu(\mathcal{V}(\Lambda_S(t))).$$

When the number of schools listed follows a Poisson distribution with mean ℓ ,

$$(58) \quad \alpha \mu(\alpha) = 1 - e^{-\alpha \ell}.$$

When schools are equally popular and have a single seat,

$$(59) \quad \mathcal{V}(\lambda) = e^{-\lambda} = 1 - \mathcal{E}(\lambda).$$

Substituting (58) and (59) into (57) yields

$$\frac{d}{dt} \mathcal{E}(\Lambda_S(t)) = -\rho(1 - e^{-\ell(1-\mathcal{E}(\Lambda_S(t)))}),$$

which (with initial condition $\mathcal{E}(\Lambda_S(1)) = 0$) has solution

$$\mathcal{E}(\Lambda_S(t)) = 1 - \frac{1}{\ell} \log(1 + (e^\ell - 1)e^{-\rho \ell(1-t)}).$$

The claimed expressions for the single lottery follows from $F_S(\infty) = \mathcal{E}(\Lambda_S(0))/\rho$ (see Lemma 5).

With independent lotteries,

$$\rho F_I(\infty) = \mathcal{E}(\Lambda_I(0)) = 1 - e^{-\Lambda_I(0)},$$

where the second equality uses (59). We can solve this for $\Lambda_I(0)$ to get

$$\Lambda_I(0) = -\log(1 - \rho F_I(\infty)),$$

from which it follows that

$$(60) \quad \mathcal{A}(\Lambda_I(0)) = \frac{\rho F_I(\infty)}{-\log(1 - \rho F_I(\infty))}$$

Furthermore, (10) implies that

$$\Lambda_I(0) = \rho \mu(\mathcal{A}(\Lambda_I(0))) = \rho \frac{\mathcal{E}(\ell \mathcal{A}(\Lambda_I(0)))}{\mathcal{A}(\Lambda_I(0))},$$

where the second equality uses (58) and (59). Multiplying each side by $\mathcal{A}(\Lambda_I(0))/\rho$ and applying Lemma 5 we get

$$F_I(\infty) = \mathcal{E}(\ell \mathcal{A}(\Lambda_I(0))) = 1 - e^{-\ell \mathcal{A}(\Lambda_I(0))},$$

from which it follows that

$$\mathcal{A}(\Lambda_I(0)) = -\frac{1}{\ell} \log(1 - F_I(\infty)).$$

Combining this with (60) yields (55). □